

Dexmedetomidine: Drug information - UpToDate

Dexmedetomidine: Drug information - UpToDate

Official reprint from UpToDate®

www.uptodate.com © 2026 UpToDate, Inc. and/or its affiliates. All Rights Reserved.

Dexmedetomidine: Drug information

2026© UpToDate, Inc. and its affiliates and/or licensors. All Rights Reserved.

Contributor Disclosures

For additional information see "Dexmedetomidine: Patient drug information" and "Dexmedetomidine: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Igalmi;
- Precedex

Brand Names: Canada

- Precedex

Pharmacologic Category

- Alpha₂-Adrenergic Agonist;
- Sedative

Dosing: Adult

Dosage guidance:

Dosing: Errors have occurred due to clinician misinterpretation of dosing information and units. Maintenance dose is expressed as mcg/kg/hour (Ref).

Clinical considerations: Does not provide adequate and reliable amnesia; therefore, use of additional agents with amnestic properties (eg, benzodiazepines, propofol) may be necessary.

Agitation associated with schizophrenia or bipolar I or II disorder

Agitation associated with schizophrenia or bipolar I or II disorder (alternative agent):

Note: Some experts recommend reserving use for patients with mild or moderate agitation who cannot tolerate first-line treatments (Ref). Monitor vital signs including orthostatic measures after each dose to prevent falls and syncope. Due to risk of hypotension, additional half-doses are not recommended in patients with systolic blood pressure (SBP) <90 mm Hg, diastolic blood pressure (DBP) <60 mm Hg, heart rate <60 beats per minute, or postural decrease in SBP ≥20 mm Hg or in DBP ≥10 mm Hg. Patients should not perform activities that require mental alertness (eg, operating machinery, driving) for at least 8 hours after administration.

Mild or moderate agitation: Sublingual film: **Sublingual/Buccal:** Initial: 120 mcg; if agitation persists, up to 2 additional doses of 60 mcg may be administered at least 2 hours apart. Maximum: 240 mcg/day.

Severe agitation: Sublingual film: **Sublingual/Buccal:** Initial: 180 mcg; if agitation persists, up to 2 additional doses of 90 mcg may be administered at least 2 hours apart. Maximum: 360 mcg/day.

General anesthesia, maintenance

General anesthesia, maintenance (adjunctive agent) (off-label use):

Note: May reduce sedative-hypnotic and/or opioid requirements of general anesthesia; may reduce postoperative opioid requirements (Ref).

Continuous **IV** infusion: Usual dosage range: 0.1 to 0.8 mcg/kg/**hour**; titrate to desired effect (Ref).

Procedural sedation or monitored anesthesia care

Procedural sedation or monitored anesthesia care (including flexible scope intubation [awake]):

IV: Initial: Loading dose of 0.5 to 1 mcg/kg over 10 minutes (use lower range for less invasive procedures [eg, ophthalmic]), followed by a continuous infusion of 0.2 to 1 mcg/kg/**hour**; titrate to desired level of sedation.

Sedation, critically ill

Sedation, critically ill:

Note: Deep sedation is not achievable with dexmedetomidine monotherapy (Ref).

IV:

Loading dose (optional): Note: Not recommended due to concerns for hemodynamic compromise (eg, hypertension, hypotension, bradycardia) (Ref).

Initial: 1 mcg/kg over 10 minutes, followed by a continuous infusion.

Continuous infusion: Usual dosage range: 0.2 to 1.5 mcg/kg/**hour**; titrate by 0.2 mcg/kg/**hour** every 30 minutes to sedation goal (eg, Richmond Agitation Sedation Scale score 0 to -2) or clinical effect (eg, ventilator synchrony) (Ref).

Note: Although infusion rates as high as 2.5 mcg/kg/**hour** have been used, doses >1.5 mcg/kg/**hour** do not provide additional clinical efficacy (Ref). Manufacturer recommends that infusion duration not exceed 24 hours; however, randomized clinical trials have demonstrated efficacy and safety up to ~14 days (Ref). Withdrawal symptoms (eg, hypertension, tachycardia, delirium, agitation) may be more likely to occur in patients with a history of hypertension or those receiving dexmedetomidine for longer durations, at greater cumulative doses (eg, >12 mcg/kg/**day**), or higher peak rates (eg, >0.8 mcg/kg/**hour**). In such patients, avoid abrupt discontinuation; carefully decrease dose, while monitoring for withdrawal symptoms (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: No dosage adjustment necessary for any degree of kidney dysfunction (Ref).

Hemodialysis, intermittent (thrice weekly): Unlikely to be dialyzed (highly protein bound): No supplemental dose or dosage adjustment necessary (Ref).

Peritoneal dialysis: Unlikely to be dialyzed (highly protein bound): No dosage adjustment necessary (Ref).

CRRT: No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Liver impairment prior to treatment initiation:

Initial or dose titration in patients with preexisting liver cirrhosis:

IV:

Note: Dexmedetomidine undergoes extensive hepatic metabolism via glucuronidation, which is highly dependent on liver blood flow. In the setting of severe liver insufficiency, clearance was reduced by ~50% (Ref). Accumulation is expected in the setting of advanced liver impairment (Ref).

Child-Turcotte-Pugh class A and B: No dosage adjustment necessary (Ref).

Child-Turcotte-Pugh class C:

General anesthesia, ICU sedation: Use lowest end of the recommended dosage range; titrate to desired effect (Ref).

Procedural sedation or monitored anesthesia care:

Loading dose: No dosage adjustment necessary (Ref).

Continuous infusion: Reduce dose to 0.1 to 0.5 mcg/kg/hour; titrate to desired level of sedation (Ref).

Sublingual film: Agitation associated with schizophrenia or bipolar I or II disorder (Ref):

Note: Monitor vital signs, including orthostatic measures, after each dose to prevent falls and syncope. Due to risk of hypotension, use of additional doses following the initial dose is not recommended in patients with systolic blood pressure (SBP) <90 mm Hg, diastolic blood pressure (DBP) <60 mm Hg, heart rate <60 beats per minute, or postural decrease in SBP ≥20 mm Hg or in DBP ≥10 mm Hg. Patients should not perform activities that require mental alertness (eg, operating machinery, driving) for at least 8 hours after administration.

Child-Turcotte-Pugh class A and B:

Mild or moderate agitation: Initial: 90 mcg; if agitation persists, up to 2 additional doses of 60 mcg may be administered at least 2 hours apart. Maximum: 210 mcg/day.

Severe agitation: Initial: 120 mcg; if agitation persists, up to 2 additional doses of 60 mcg may be administered at least 2 hours apart. Maximum: 240 mcg/day.

Child-Turcotte-Pugh class C:

Mild or moderate agitation: Initial: 60 mcg; if agitation persists, up to 2 additional doses of 60 mcg may be administered at least 2 hours apart. Maximum: 180 mcg/day.

Severe agitation: Initial: 90 mcg; if agitation persists, up to 2 additional doses of 60 mcg may be administered at least 2 hours apart. Maximum: 210 mcg/day.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, or 3 obesity (BMI ≥30 kg/m²):

IV: Use **adjusted body weight** for initial weight-based dose calculations, then titrate to clinical effect (Ref). Clinicians should **not** change dosing weight from one weight metric to another during therapy (ie, adjusted body weight to/from actual body weight) (Ref). Refer to adult dosing for indication-specific doses.

Rationale for recommendations:

There are limited data on the effects of obesity on dexmedetomidine dosing or pharmacokinetics. Pharmacokinetic studies evaluating dexmedetomidine, a lipophilic compound, have shown a meaningful correlation between both fat free mass (FFM) and lean body weight (LBW) with clearance, but not between actual body weight and clearance (Ref). For V_d , defining the most descriptive weight metric is less clear, but dexmedetomidine appears to have a small additive volume above FFM and LBW (Ref).

Some data suggest using LBW for initial weight-based dosing of dexmedetomidine (Ref); however, there are concerns with calculation errors when using the complex LBW and FFM formula, especially in critical situations (Ref). In addition, if LBW is used, underdosing may occur in clinical situations where faster sedation is indicated (eg, procedural sedation, induction of anesthesia, ICU sedation) (Ref). Using actual body weight in weight-based dosing calculations for bolus doses or continuous infusions may lead to excessive concentrations resulting in adverse effects (Ref). Use of an alternative size descriptor (ie, adjusted body weight) is recommended for dosing in patients with class 1, 2, or 3 obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$) (Ref).

Dosing: Older Adult

Agitation associated with schizophrenia or bipolar I or II disorder:

Note: Monitor vital signs including orthostatic measures after each dose to prevent falls and syncope. Due to risk of hypotension, additional half-doses are not recommended in patients with systolic blood pressure (SBP) $<90 \text{ mm Hg}$, diastolic blood pressure (DBP) $<60 \text{ mm Hg}$, heart rate <60 beats per minute, or postural decrease in SBP $\geq 20 \text{ mm Hg}$ or in DBP $\geq 10 \text{ mm Hg}$. Patients should not perform activities that require mental alertness (eg, operating machinery, driving) for at least 8 hours after administration.

Mild, moderate, or severe agitation: Sublingual film: **Sublingual/Buccal:** Initial: 120 mcg; if agitation persists, up to 2 additional doses of 60 mcg may be administered at least 2 hours apart. Maximum: 240 mcg/day.

Mechanically ventilated patients in the ICU, sedation: IV: Refer to adult dosing. Consider dosage reduction. No specific guidelines available. Dose selections should be cautious, at the low end of dosage range; titration should be slower, allowing adequate time to evaluate response.

Procedural sedation or monitored anesthesia care (including flexible scope intubation [awake]): IV: Refer to adult dosing: Initial: Loading dose of 0.5 mcg/kg over 10 minutes; Maintenance infusion: Dosage reduction should be considered.

Dosing: Pediatric

(For additional information see "Dexmedetomidine: Pediatric drug information")

Dosage guidance:

Dosing: Errors have occurred due to misinterpretation of dosing information. Maintenance dose expressed as mcg/kg/ **hour**. Individualize and titrate to desired clinical effect.

Safety: Consult local regulations and individual institutional policies and procedures; should only be used by experienced personnel who are not actively engaged in the procedure or surgery.

ICU sedation

ICU sedation: Limited data available:

Infants, Children, and Adolescents:

Loading dose (Optional): IV: 0.5 to 1 mcg/kg/dose over 10 minutes; use of loading dose is dependent upon concomitant sedation agents and patient's current and desired level of sedation (Ref).

Maintenance dose: Note: Although the manufacturer recommends limiting continuous infusions to ≤ 24 hours in adult patients, duration in pediatric patients has been reported from 2 hours to 103 days; weaning of dexmedetomidine and/or replacement strategies have been recommended (Ref).

Continuous IV infusion: Initial: 0.2 to 0.5 mcg/kg/hour; increase dose by 0.1 to 0.3 mcg/kg/hour to achieve desired level of sedation; usual dose range: 0.2 to 2.5 mcg/kg/hour (Ref). **Note:** Infants may require higher maintenance infusion rates than either neonates or older children (Ref).

Junctional ectopic tachycardia, postoperative; prevention

Junctional ectopic tachycardia, postoperative; prevention : Limited data available; dosing regimens variable; optimal dose not established:

Infants, Children, and Adolescents (very limited data for adolescents) (Ref):

Loading dose: 0.5 to 1 mcg/kg as a single dose over 15 to 20 minutes; infusion should be completed 10 minutes prior to anesthesia induction.

Continuous IV infusion: 0.5 to 0.75 mcg/kg/hour; administer immediately following loading dose and continue intraoperatively and postoperatively for 48 hours.

Sedation, noninvasive procedures; nonintubated

Sedation, noninvasive procedures; nonintubated (nonpainful or minimally painful):

Intranasal: Limited data available; reported dosing regimens variable and ideal dose not established:

Infants and Children (very limited data in children >10 years): Intranasal: Usual dose: 2 to 3 mcg/kg as a single dose 30 to 60 minutes prior to procedure; reported dose range: 1 to 4 mcg/kg; dosing based on multiple studies evaluating intranasal dexmedetomidine administered prior to procedures (eg, CT, MRI, transesophageal echocardiography, ophthalmic exams, auditory brainstem response [brainstem auditory evoked response] test) (Ref).

IM: Limited data available; reported dosing regimens variable and ideal dose not established:

Infants, Children, and Adolescents: IM: Initial dose: 1 to 4 mcg/kg as a single dose prior to procedure; if sedation inadequate, may give a second dose of 1 to 2 mcg/kg; dosing based on multiple studies evaluating intramuscular dexmedetomidine administered prior to procedures (eg, CT, MRI, EEG) (Ref).

IV:

Infants, Children, and Adolescents:

Loading dose: IV: Usual dose: 1 to 2 mcg/kg/dose over 10 minutes; may repeat if sedation is inadequate; reported range: 0.5 to 3 mcg/kg/dose (Ref). **Note:** A lower loading dose may be needed when administered with concomitant anesthetics, sedatives, hypnotics, or opioids (eg, 0.5 to 1 mcg/kg) (Ref).

Maintenance dose: Continuous IV infusion: Usual dose range: 1 to 2 mcg/kg/hour; titrate to achieve clinical effect; reported range: 0.5 to 2 mcg/kg/hour (Ref). **Note:** Lower doses have been reported when administered concomitantly with midazolam or propofol (Ref).

Sedation, pre-anesthetic

Sedation, pre-anesthetic: Limited data available:

Intranasal:

Children and Adolescents (very limited data available in patients >9 years): Intranasal: 1 to 2 mcg/kg as a single dose 30 to 60 minutes prior to induction of anesthesia. Higher-end doses (2 mcg/kg) are recommended for older children (≥5 years) and adolescents (Ref).

Oral: Dosing regimens variable; optimal dose not established:

Children and Adolescents (very limited data available in adolescents): Oral: 1 to 4 mcg/kg as a single dose 45 minutes prior to induction of anesthesia; dosing based on three retrospective studies and one prospective study (Ref). **Note:** A dose of 1 mcg/kg has not been shown to decrease the incidence of emergence delirium (Ref).

Dosage adjustment for extracorporeal life support/extracorporeal membrane oxygenation (ECLS/ECMO):

ECLS/ECMO considerations:

Components of ECLS/ECMO circuit and pathophysiological changes during critical illness influence the pharmacokinetic/pharmacodynamic (PK/PD) parameters (eg, protein binding, volume of distribution [V_d], clearance, partition coefficient, drug exposure, target attainment goal) of a drug and overall drug exposure in critically ill patients; ECLS/ECMO circuit components include type and age of oxygenator and tubing (size, length, and coating), blood flow rate through the circuit, and the addition of dialysis; pathophysiologic factors include organ function (kidney and liver), volume status, and serum albumin concentration (Ref). Significant interpatient variability has been described (Ref); monitor patient response closely; refer to institutional protocols.

Administration of medication using the patient's vascular access is preferred; if patient's vascular access is unavailable, administration of medication into ECLS/ECMO circuit is an acceptable alternative (Ref).

Sedation: Very limited data available; ideal dose and dose-response not established:

Infants, Children, and Adolescents: IV: Initial: There are no dosage recommendations available in the literature. Based on drug characteristics (high protein binding and lipophilicity), removal by the ECLS/ECMO circuit, and potential increase in hepatic clearance, initiate a continuous IV infusion and monitor closely; adjust dose based on patient-specific factors (eg, organ function, clinical response, and sedation goals) (Ref); consult institutional policies.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Parenteral: There are no dosage adjustments provided in the manufacturer's labeling; however, pharmacokinetics were not significantly different in adult patients with severe renal impairment (CrCl <30 mL/minute).

Dosing: Liver Impairment: Pediatric

Parenteral: There are no pediatric-specific dosage adjustments provided in the manufacturer's labeling; however, clearance is reduced in varying degrees based on the level of hepatic impairment; adult data suggest a dosage adjustment.

Adverse Reactions (Significant): Considerations

Cardiovascular effects

Dexmedetomidine has been associated with **hypotension** and/or **bradycardia** (Ref). **Hypertension**, although less common, may also occur (Ref). Bradycardia, hypotension, and hypertension have occurred in pediatric patients (Ref). Dexmedetomidine has been associated with cardiac arrest/asystole in case reports (Ref). Bradycardia, hypotension, and hypertension are all more common with the IV formulation.

Mechanism: Dexmedetomidine can cause a biphasic blood pressure response (a short hypertensive phase with hypotension thereafter). These effects are related to alpha-2 adrenergic receptor subtypes: Alpha-2B (responsible for hypertensive effects) and alpha-2A (responsible for hypotension) (Ref). In addition, dexmedetomidine decreases sinus and atrioventricular nodal function in pediatric patients (Ref).

Onset: Hypotension and/or bradycardia: Rapid. In one study, the median time to first hypotensive episode was 5 hours (range: ~3 to 13 hours) (Ref). Another study reported a median time of ~4 hours for hemodynamic instability, with more than two-thirds of patients developing hypotension and/or bradycardia within 24 hours (Ref). An additional study showed a median onset of 17 hours for the first hemodynamic event (hypotension or bradycardia) (Ref).

Risk factors:

- Hypotension and/or bradycardia:
- Low baseline arterial blood pressure (Ref)

- Increased age (Ref)
- Increased severity of illness (Ref)
- Loading dose (risk for hypotension in neurosurgical patients or risk for bradycardia in general ICU patients) (Ref)
- Inotrope use (cardiac ICU patients) (Ref)
- Coronary artery disease (Ref)
- Lack of adjuvant sedatives (Ref)
- Titration of infusion < every 30 minutes (Ref)
- Infusion >6 hours (pediatric patients) (Ref)
- Hypertension:
 - Loading dose and higher peak plasma concentrations (risk for transient hypertension) (Ref)
- Cardiac arrest/asystole:
 - Bradycardia or acute hypotension (Ref)

Tolerance, tachyphylaxis, and withdrawal

Prolonged use of dexmedetomidine use may lead to **drug tolerance**, **tachyphylaxis**, loss of sedation control, increased adverse reactions, and **withdrawal syndrome** (Ref). Withdrawal symptoms may occur in both adult and pediatric patients and may include hypertension, tachycardia, diaphoresis, anxiety, fever, and delirium (Ref). The incidence of withdrawal has been reported in ~30% of patients (Ref).

Mechanism: Dose- and time-related; related to alpha-2 receptor affinity, binding, and upregulation after prolonged use (Ref).

Onset: Rapid; withdrawal symptoms typically occur with prolonged use (eg, >24 hours) (Ref). However, mild, transient withdrawal symptoms (eg, agitation, delirium) have occurred with short term use (<2 hours) in pediatric patients.

Risk factors:

- Higher cumulative daily doses (eg, >12 mcg/kg/day) or with higher peak rates (eg, >0.8 mcg/kg/hour) (Ref)
- Prolonged use (Ref)
- History of hypertension (Ref)
- Duration of concurrent opioid prior to dexmedetomidine discontinuation (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Frequency dependent upon dose, duration, and indication. Reported adverse reactions are for IV dexmedetomidine in adults unless otherwise noted.

>10%:

Cardiovascular: Bradycardia (infants, children, and adolescents: IV: 57% to 71%; adults: IV: 5% to 42%, sublingual: 2%;) (table 1) Bradycardia, hypertension (infants, children, and adolescents: IV: 26% to 47%; adults: IV: 11% to 16%, sublingual: 5%) (table 2) Hypertension, hypotension (infants, children, and adolescents: IV: 16% to 31%; adults: IV: 24% to 56%, sublingual: 5%) (table 3) Hypotension, tachycardia (infants, children, and adolescents: IV: 2% to 7%; adults: 25%)

Drug (Dexmedetomidine)	Comparator	Placebo	Population	Dosage Form
71%	N/A	N/A	Infants, children, and adolescents	IV: High dose
57%	N/A	N/A	Infants, children, and adolescents	IV: Low dose
57%	N/A	N/A	Infants, children, and adolescents	IV: Middle dose
2%	N/A	0%	Adults	Sublingual film
2%	N/A	0%	Adults	Sublingual film
7%	N/A	3%	Adults	IV
5%	Propofol: 0%	3%	Adults	IV: 0.5 mcg/kg
42%	Midazolam: 19%	N/A	Adults	IV
14%	N/A	4%	Adults	IV: 1.3 mcg/kg

Drug (Dexmedetomidine)	Comparator	Placebo	Population	Dosage Form
47%	N/A	N/A	Infants, children, and adolescents	IV: High dose
41%	N/A	N/A	Infants, children, and adolescents	IV: Middle dose
26%	N/A	N/A	Infants, children, and adolescents	IV: Low dose
16%	N/A	18%	Adults	IV
13%	Propofol: 4%	19%	Adults	IV: 0.5 mcg/kg
11%	Midazolam: 15%	N/A	Adults	IV
13%	N/A	24%	Adults	IV: 1.3 mcg/kg

Drug (Dexmedetomidine)	Comparator	Placebo	Population	Dosage Form
31%	N/A	N/A	Infants, children, and adolescents	IV: Low dose
26%	N/A	N/A	Infants, children, and adolescents	IV: Middle dose
16%	N/A	N/A	Infants, children, and adolescents	IV: High dose
5%	N/A	0%	Adults	Sublingual film
5%	N/A	0%	Adults	Sublingual film
28%	N/A	13%	Adults	IV
24%	Propofol: 13%	12%	Adults	IV: 0.5 mcg/kg
56%	Midazolam: 56%	N/A	Adults	IV
54%	N/A	30%	Adults	IV: 1.3 mcg/kg

Gastrointestinal: Constipation (6% to 14%), nausea (IV: 3% to 11%; sublingual: 3%)

Nervous system: Agitation (5% to 14%), drowsiness (including fatigue and lethargy: sublingual: 22% to 23%)

Respiratory: Bradypnea (infants, children, and adolescents: 58% to 79%), hypoxia (infants, children, and adolescents: 3% to 14%), respiratory depression (37%)

1% to 10%:

Cardiovascular: Atrial fibrillation (2% to 9%), edema (2%), orthostatic hypotension (sublingual: 3% to 5%), peripheral edema (3% to 7%)

Endocrine & metabolic: Hyperglycemia (7%), hypocalcemia (1%), hypoglycemia (5%), hypokalemia (9%), hypomagnesemia (1%), hypovolemia (3%), increased thirst (2%)

Gastrointestinal: Abdominal distress (sublingual: 2%), oral hypoesthesia (sublingual: $\leq 7\%$), xerostomia (IV: 3%; sublingual: 4% to 7%)

Genitourinary: Decreased urine output (1%), oliguria (2%)

Hematologic & oncologic: Anemia (3%)

Nervous system: Anxiety (5% to 9%), dizziness (sublingual: 4% to 6%), paresthesia (sublingual: $\leq 7\%$), withdrawal syndrome (ICU sedation; infants, children, and adolescents: 2%; adults: 3% to 5%)

Renal: Acute kidney injury (2% to 3%)

Respiratory: Acute respiratory distress syndrome (1% to 9%), pleural effusion (2%), respiratory failure (2% to 10%), wheezing (1%)

Miscellaneous: Fever (5% to 7%)

Postmarketing:

Cardiovascular: Acute myocardial infarction, atrioventricular block, cardiac arrhythmia, extrasystoles, inversion T wave on ECG, prolonged QT interval on ECG (IV, sublingual), sinoatrial arrest, supraventricular tachycardia, ventricular arrhythmia, ventricular tachycardia

Dermatologic: Hyperhidrosis, pruritus, skin rash, urticaria

Endocrine & metabolic: Acidosis, hyperkalemia, hypernatremia, respiratory acidosis

Gastrointestinal: Abdominal pain, diarrhea, gastrointestinal pseudo-obstruction (Alkaissi 2021), vomiting

Hematologic & oncologic: Hemorrhage

Hepatic: Abnormal liver function, hyperbilirubinemia, increased gamma-glutamyl transferase, increased serum alanine aminotransferase, increased serum alkaline phosphatase, increased serum aspartate aminotransferase

Nervous system: Anesthesia (light), chills, confusion, delirium, hallucination, headache, hyperpyrexia, illusion, neuralgia, neuritis, pain, seizure, speech disturbance

Ophthalmic: Photopsia, visual disturbance

Renal: Increased blood urea nitrogen, polyuria (Zhu 2022)

Respiratory: Apnea, bronchospasm, dyspnea, hypercapnia, hypoventilation, hypoxia, pulmonary congestion

Miscellaneous: Drug tolerance (use >24 hours) (Tobias 2010), tachyphylaxis (use >24 hours)

Contraindications

There are no contraindications listed in the US manufacturer's labeling.

Canadian labeling: Hypersensitivity to dexmedetomidine or any component of the formulation.

Warnings/Precautions

Disease-related concerns:

- Cardiovascular disease: Use IV formulation with caution in patients with heart block, bradycardia, severe ventricular dysfunction, hypovolemia, or chronic hypertension. In a scientific statement from the American Heart Association, dexmedetomidine has been determined to be an agent that may exacerbate underlying myocardial dysfunction (magnitude: moderate) (AHA [Page 2016]). Avoid use of the sublingual film in patients with hypotension, orthostatic hypotension, advanced heart block, severe ventricular dysfunction, history of syncope or other arrhythmias, symptomatic bradycardia,

hypokalemia or hypomagnesemia, in patients at risk of torsades de pointes or sudden death, including those with known QT prolongation, and in patients receiving other drugs known to prolong the QT interval.

- Diabetes: Use with caution in patients with diabetes mellitus; cardiovascular adverse events (eg, bradycardia, hypotension) may be more pronounced.
- Hemodynamics: Bradycardia and/or hypotension may occur with loading doses. Transient, paradoxical hypertension may also occur with loading doses (ie, high peak plasma concentrations during rapid loading). Higher maintenance doses may cause bradycardia and hypertension, although hypotension may also occur (during titration or with dose escalations more frequently than every 30 minutes) (Ebert 2000; Gerlach 2009; Gerlach 2016; Tan 2010; Zhang 2016).
- Liver impairment: Use with caution in patients with liver impairment; dosage adjustment may be needed.

Special populations:

- Older adults: Use with caution in older adults; cardiovascular events (eg, bradycardia, hypotension) may be more pronounced. Dose reduction may be necessary.

Other warnings/precautions:

- Arousability: Patients may be arousable and alert when stimulated with use of the IV formulation. This alone should not be considered as lack of efficacy in the absence of other clinical signs/symptoms.
- Experienced personnel: IV formulation should be administered only by persons skilled in management of patients in intensive care setting or operating room. Patients should be continuously monitored.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Film, Sublingual, as hydrochloride:

Igalmi: 120 mcg (10 ea); 180 mcg (10 ea) [contains fd&c blue #1 (brilliant blue)]

Solution, Intravenous:

Generic: 400 mcg/4 mL (4 mL); 1000 mcg/10 mL (10 mL)

Solution, Intravenous, as hydrochloride:

Precedex: Dexmedetomidine 200 mcg/50 mL in NaCl 0.9% (50 mL); Dexmedetomidine 400 mcg/100 mL in NaCl 0.9% (100 mL)

Generic: 200 mcg/2 mL (2 mL); Dexmedetomidine 200 mcg/50 mL in NaCl 0.9% (50 mL); Dexmedetomidine 400 mcg/100 mL in NaCl 0.9% (100 mL)

Solution, Intravenous, as hydrochloride [preservative free]:

Precedex: 1000 mcg/250 mL (250 mL); 200 mcg/2 mL (2 mL); Dexmedetomidine 400 mcg/100 mL in NaCl 0.9% (100 mL [DSC]) [additive free, latex free]

Precedex: 80 mcg/20 mL (20 mL); Dexmedetomidine 200 mcg/50 mL in NaCl 0.9% (50 mL); Dexmedetomidine 400 mcg/100 mL in NaCl 0.9% (100 mL) [latex free]

Precedex: 80 mcg/20 mL (20 mL); 1000 mcg/250 mL (250 mL); Dexmedetomidine 200 mcg/50 mL in NaCl 0.9% (50 mL); Dexmedetomidine 400 mcg/100 mL in NaCl 0.9% (100 mL)

Generic: 80 mcg/20 mL (20 mL); 1000 mcg/250 mL (250 mL); 200 mcg/2 mL (2 mL); Dexmedetomidine 200 mcg/50 mL in Dextrose 5% (50 mL); Dexmedetomidine 200 mcg/50 mL in NaCl 0.9% (50 mL); Dexmedetomidine 400 mcg/100 mL in Dextrose 5% (100 mL); Dexmedetomidine 400 mcg/100 mL in NaCl 0.9% (100 mL)

Generic Equivalent Available: US

May be product dependent

Pricing: US

Film (Igalmi Sublingual)

120 mcg (per each): \$126.00

180 mcg (per each): \$126.00

Solution (dexmedetomidine HCl in NaCl Intravenous)

80 mcg/20 mL (per mL): \$1.63 - \$1.71

200 mcg/50 mL (per mL): \$0.28 - \$1.08

400 mcg/100 mL (per mL): \$0.22 - \$1.00

1000MCG/250ML (per mL): \$0.24

Solution (dexmedetomidine HCl Intravenous)

200 mcg/2 mL (per mL): \$2.02 - \$38.52

400MCG/4ML (per mL): \$21.38 - \$23.69

1000MCG/10ML (per mL): \$14.40 - \$15.99

Solution (dexmedetomidine HCl-Dextrose Intravenous)

200MCG/50ML -5% (per mL): \$0.46

400MCG/100ML -5% (per mL): \$0.43

Solution (Precedex Intravenous)

80 mcg/20 mL (per mL): \$0.66

200 mcg/2 mL (per mL): \$3.60

200 mcg/50 mL (per mL): \$0.26

400 mcg/100 mL (per mL): \$0.23

1000MCG/250ML (per mL): \$1.02

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous:

Precedex: 4 mcg/mL (20 mL, 50 mL, 100 mL)

Generic: 4 mcg/mL (50 mL, 100 mL, 250 mL)

Solution, Intravenous, as hydrochloride:

Precedex: 200 mcg/2 mL ([DSC])

Generic: 200 mcg/2 mL (2 mL)

Administration: Adult

IV: Administer using a controlled infusion device. Advisable to use administration components made with synthetic or coated natural rubber gaskets. If loading dose used, administer over 10 minutes; may extend to 20 minutes to further reduce vasoconstrictive effects. Titration no more frequently than every 30 minutes may reduce the incidence of hypotension when used for ICU sedation (Ref).

Sublingual film: For sublingual or buccal administration. Keep in foil pouch until ready to administer; administer immediately after pouch is open and dose is prepared. Remove from pouch with clean, dry hands. For half-doses, cut film in half between the dots with clean, dry scissors; discard the unused half in a waste container. For 60 mcg dose, cut 120 mcg film in half; for 90 mcg dose, cut 180 mcg film in half. For sublingual administration, place film under the tongue; for buccal administration place film behind the lower lip. Allow the film to dissolve; do not chew or swallow. Do not eat or drink for at least 15 minutes after sublingual administration, or at least one hour after buccal administration. Monitor vital signs. Patients should sit or lie down until vital signs are within normal range; if unable to remain seated or lying down, precautions should be taken to reduce the risks of falls. Ensure patient is alert and not experiencing symptomatic hypotension or orthostatic hypotension before resuming ambulation.

Preparation for Administration: Adult

Note: Premixed solutions are also available.

Concentrated solution (100 mcg/mL): Must dilute in NS to achieve the required concentration (4 mcg/mL) prior to administration. Add 2 mL (200 mcg) of dexmedetomidine to 48 mL of NS for a total volume of 50 mL (4 mcg/mL) or 4 mL (400 mcg) of dexmedetomidine to 96 mL of NS for a total volume of 100 mL. Shake gently to mix. Do not use if solution is discolored or precipitate matter is present.

Usual Infusion Concentrations: Adult

IV infusion: 200 mcg in 50 mL (concentration: 4 mcg/mL) of NS

Administration: Pediatric

Parenteral:

IM: Administer undiluted (4 mcg/mL or 100 mcg/mL) into the deltoid or gluteal muscle (Ref).

IV: Concentrated solution (100 mcg/mL) must be diluted prior to administration; premixed IV solution (4 mcg/mL) is available. Administer using a controlled infusion device. Infuse loading dose over 10 minutes; may extend up to 20 minutes in neonatal patients or when needed to further reduce vasoconstrictive effects; rapid infusions are associated with severe side effects (Ref). Dexmedetomidine may adhere to natural rubber; use administration components made with synthetic or coated natural rubber gaskets.

Intranasal: Administer undiluted (100 mcg/mL) or dilute in a small volume of NS (eg, to a total volume of 1 or 1.5 mL). Divide dose and give half in each nostril by slowly dripping from a needleless syringe onto the nasal mucosa while in a recumbent position (Ref). Some recommend using a nasal atomizer such as the MAD Nasal Drug delivery device (Ref).

Oral: Administer parenteral formulation undiluted (100 mcg/mL) or diluted in ~5 mL of apple juice or honey (avoid use of honey in infants) (Ref).

Preparation for Administration: Pediatric

Parenteral: IV: Dexmedetomidine injection concentrate (100 mcg/mL) must be diluted prior to administration. Dilute 200 mcg (2 mL) in 48 mL NS to achieve a final concentration of 4 mcg/mL per the manufacturer; shake gently to mix; a premixed 4 mcg/mL IV solution is commercially available. Concentrations of 8 mcg/mL, 12 mcg/mL, and 20 mcg/mL have been shown to be stable (clear and without visible particulates) for 48 hours at 21°C to 25°C in PVC bags; a slight decrease in pH was

seen with increasing concentrations (Ref). ASHP recommends a standard concentration of 4 mcg/mL (Ref).

Usual Infusion Concentrations: Pediatric

Note: Premixed solutions available.

IV infusion: 4 mcg/mL, 12 mcg/mL, 20 mcg/mL.

Storage/Stability

Injection:

Premixed solution: Store at 20°C to 25°C (68°F to 77°F); excursions permitted 15°C to 30°C (59°F to 86°F). Do not freeze. Keep infusion bags in the overwrap until ready to use to protect from light.

Vials: Store unopened vials (single-dose and multi-dose) at room temperature. Diluted solutions using multi-dose vials may be stored for up to 4 hours at room temperature or up to 24 hours at 2°C to 8°C (35°F to 46°F) prior to use.

Sublingual film: Store at 20°C to 25°C (68°F to 77°F); excursions permitted from 15°C to 30°C (59°F to 86°F). Keep in the foil pouch until ready to administer.

Use: Labeled Indications

Agitation associated with schizophrenia or bipolar I or II disorder: Treatment of acute agitation associated with schizophrenia or bipolar I or II disorder in adults.

Limitations of use: Safety and effectiveness have not been established beyond 24 hours from the first dose.

Procedural sedation or monitored anesthesia care (including flexible scope intubation [awake]): Procedural sedation in adults prior to and/or during awake fiberoptic intubation; sedation prior to and/or during surgical or other procedures of nonintubated adults; sedation prior to and/or during noninvasive procedures of nonintubated pediatric patients ≥1 month of age.

Sedation, critically ill: Sedation of intubated, mechanically ventilated adults in the ICU; sedation in adults during extubation and post extubation in the ICU.

Use: Off-Label: Adult

General anesthesia, maintenance

Medication Safety Issues

Sound-alike/look-alike issues:

Dexamethasone may be confused with dexamethasone.

Precedex may be confused with Peridex.

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (moderate sedation agent, IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Administration issues:

Errors have occurred due to misinterpretation of dosing information; use caution. Maintenance dose expressed as mcg/kg/hour.

Metabolism/Transport Effects

Substrate of CYP2A6 (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Alcohol (Ethyl): CNS Depressants may increase CNS depressant effects of Alcohol (Ethyl). *Risk C: Monitor*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amisulpride (Oral): May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Androgens: Hypertension-Associated Agents may increase hypertensive effects of Androgens. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Beta-Blockers: May increase rebound hypertensive effects of Alpha2-Agonists. This effect can occur when the Alpha2-Agonist is abruptly withdrawn. Alpha2-Agonists may increase AV-blocking effects of Beta-Blockers. Sinus node dysfunction may also be enhanced. Management: Closely monitor heart rate during treatment with a beta blocker and clonidine. Withdraw beta blockers several days before clonidine withdrawal when possible, and monitor blood pressure closely. Recommendations for other alpha2-agonists are unavailable. *Risk D: Consider Therapy Modification*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ceritinib: Bradycardia-Causing Agents may increase bradycardic effects of Ceritinib. Management: If this combination cannot be avoided, monitor patients for evidence of symptomatic bradycardia, and closely monitor blood pressure and heart rate during therapy. *Risk D: Consider Therapy Modification*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

CNS Depressants: May increase CNS depressant effects of Dexmedetomidine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration

cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

DroPERidol: May increase CNS depressant effects of CNS Depressants. Management: Consider dose reductions of droperidol or of other CNS agents (eg, opioids, barbiturates) with concomitant use. *Risk D: Consider Therapy Modification*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

HydroXYzine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Levoketoconazole: QT-prolonging Agents (Indeterminate Risk - Avoid) may increase QTc-prolonging effects of Levoketoconazole. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Loxapine: CNS Depressants may increase CNS depressant effects of Loxapine. Management: Consider reducing the dose of CNS depressants administered concomitantly with loxapine due to an increased risk of respiratory depression, sedation, hypotension, and syncope. *Risk D: Consider Therapy Modification*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methotrimeprazine: CNS Depressants may increase CNS depressant effects of Methotrimeprazine. Methotrimeprazine may increase CNS depressant effects of CNS Depressants. Management: Reduce the usual dose of CNS depressants by 50% if starting methotrimeprazine until the dose of methotrimeprazine is stable. Monitor patient closely for evidence of CNS depression. *Risk D: Consider Therapy Modification*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirtazapine: May decrease therapeutic effects of Alpha2-Agonists. CNS depressant effects of Alpha2-Agonists and Mirtazapine may increase with coadministration. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to

obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): CNS Depressants may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). Management: Consider alternatives to this combination when possible. If combined, dose reduction or discontinuation of one or more CNS depressants (including the oxybate salt product) should be considered. Interrupt oxybate salt treatment during short-term opioid use *Risk D: Consider Therapy Modification*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

QT-prolonging Agents (Highest Risk): QT-prolonging Agents (Indeterminate Risk - Avoid) may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management:

Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIrole: CNS Depressants may increase sedative effects of ROPINIrole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Scopolamine: CNS Depressants may increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Serotonin/Norepinephrine Reuptake Inhibitor: May decrease therapeutic effects of Alpha2-Agonists. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Solriamfetol: May increase hypertensive effects of Hypertension-Associated Agents. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

TiZANidine: Alpha2-Agonists may increase hypotensive effects of TiZANidine. *Risk X: Avoid*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Dexmedetomidine crosses the placenta.

Placental transfer was demonstrated following IV infusion in patients undergoing cesarean delivery (Eskandr 2018; Wang 2017; Yu 2015). Based on data from human placental perfusion studies,

dexmedetomidine may be retained in the placenta, partially limiting the total amount that could be transferred to the fetus (Ala-Kokko 1997).

Bradycardia is reported with dexmedetomidine use. Fetal bradycardia was noted following a maternal IV bolus dose administered during surgery to a patient on dexmedetomidine infusion for a percutaneous balloon mitral valvuloplasty at 27 weeks' gestation (Weiner 2014).

IV and intrathecal dexmedetomidine have been evaluated for use during spinal anesthesia in patients undergoing cesarean delivery (Bao 2017; Liu 2019; Shen 2020; Sun 2020; Wang 2019). Data from available meta-analyses of randomized controlled studies found no adverse effects of exposure on 1- and 5-minute Apgar scores or umbilical cord blood gases (Lee 2021; Shen 2020; Sun 2020; Zhang 2017). Decreased maternal shivering, nausea, and vomiting have been observed in some studies; dexmedetomidine may also improve the onset to anesthesia compared to some other agents (Bao 2017; Liu 2019; Shen 2020; Sun 2020; Wang 2019; Zhang 2017).

Dexmedetomidine (in combination with other agents) has also been evaluated for use in patient-controlled analgesia following cesarean delivery (Liu 2021; Nie 2018; Zhang 2021).

Breastfeeding Considerations

Dexmedetomidine is present in breast milk.

- A study to test a method of detecting dexmedetomidine in breast milk presented data collected following administration to 4 women undergoing cesarean delivery. Dexmedetomidine IV was given at 6 mcg/kg/hour for 10 minutes, followed by 0.2 to 0.7 mcg/kg/hour until closure of the peritoneal incision. Dexmedetomidine was present in maternal serum of all patients at the end of the infusion (range: 390.3 to 744 pg/mL). Breast milk and maternal serum were sampled 6 and 24 hours after the infusion stopped. Breast milk concentrations at 6 hours ranged from less than the level of detection (<3 pg/mL) to 30.4 pg/mL and all breast milk samples obtained at 24 hours were less than the level of detection. Using the data collected at 6 hours, authors of the study calculated the relative infant dose (RID) of dexmedetomidine to be <1% of the weight-adjusted maternal dose (range: 0.040% to 0.098%) (Nakanishi 2017).
- Dexmedetomidine was evaluated in the colostrum of 10 patients following cesarean delivery. Patients received dexmedetomidine 6 mcg/kg/hour over 10 minutes following removal of the neonate followed by a maintenance dose of 0.7 mcg/kg/hour until closure of the peritoneal incision (mean dose: 63.9 mcg over 21 ± 5 minutes). Maternal serum levels at the end of the infusion ranged from 293.5 to 744 pg/mL. Six hours following dexmedetomidine administration, breast milk concentrations ranged from below the limit of detection to 30.4 pg/mL. Only 1 sample had detectable concentrations at 12 hours (13.6 pg/mL) and dexmedetomidine was not measurable in any breast milk sample 24 hours after the dose. Authors of this study calculated the RID at 6 hours to be <1% of the weight-adjusted maternal dose (range: 0.023% to 0.098%). Breastfeeding was adequate for all neonates at 3 and 5 days, except for 1 infant at 3 days (Yoshimura 2017).
- The presence of dexmedetomidine in breast milk was evaluated in patients receiving patient-controlled IV analgesia (PCIA) following cesarean delivery. Following a bolus dose of 0.5 mcg/kg, pumps were programmed to deliver dexmedetomidine 0.03 mcg/kg/hour (n = 29), 0.05 mcg/kg/hour (n = 29), or 0.08 mcg/kg/hour (n = 28) in combination with butorphanol. The basal infusion rate and bolus doses were the same for each study group. Breast milk was collected for 48 hours once lactation started. Dexmedetomidine was present at all doses evaluated. The highest concentrations were observed in patients receiving 0.08 mcg/kg/hour. Actual concentrations were not presented; however, authors of the study calculated the RID of dexmedetomidine to be <1% of the weight-adjusted maternal dose. In addition, the 0.05 mcg/kg/hour dose was found to be most effective for maternal analgesia (Liu 2021).
- A case report describes dexmedetomidine breast milk concentrations in a patient who underwent an awake left frontal craniotomy at 4 weeks postpartum. Dexmedetomidine 45 mcg was administered as a bolus injection followed by an infusion of 0.7 to 1 mcg/kg/hour during surgery. Breast milk was collected twice during surgery and twice after. Concentrations of dexmedetomidine in breast milk were 88 pg/mL ~2 hours after surgery was initiated, and 50 pg/mL ~4 hours into surgery (following a pause in the infusion). Following surgery, breast milk samples obtained in the recovery

room were 89 pg/mL (48 minutes after discontinuation of infusion) and 15 pg/mL (4 hours after the dexmedetomidine infusion was stopped) (Dodd 2021).

- In general, breastfeeding is considered acceptable when the RID of a medication is <10% (Anderson 2016; Ito 2000).

A randomized, prospective, double-blind study was conducted to evaluate the effect of dexmedetomidine on breastfeeding outcomes. Patients who planned to exclusively breastfeed for ≥ 3 months following an elective cesarean delivery were eligible. Dexmedetomidine was initiated following cutting of the umbilical cord with a loading dose of 0.5 mc/kg, followed by 0.5 mcg/kg/hour until surgery ended. Patients were then connected to a PCIA pump and given dexmedetomidine 2 mcg/kg in combination with sufentanil and dolasetron ($n = 72$). Outcomes were compared to patients receiving similar care, with the exception of dexmedetomidine ($n = 72$). The mean GA at delivery was 39.34 weeks and 39.03 for the dexmedetomidine and standard care groups, respectively ($p = 0.038$). The 1- and 5-minute Apgar scores and duration of surgery were similar for both groups. The time to first lactation was shorter and the probability of exclusively breastfeeding within 6 weeks postpartum was greater in the dexmedetomidine group. Milk volumes were similar on the first day postpartum, but greater on the second day in patients in the dexmedetomidine group. A neonatal behavioral neurological assessment conducted on days 1 and 2 of life showed no differences in neonatal outcomes (Wang 2020).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. Breastfeeding infants should be monitored for adverse events (eg, irritability), particularly following high maternal doses used for prolonged periods of time (Mitchell 2020; Mitchell 2021).

Monitoring Parameters

Level of sedation; heart rate, respiration, rhythm, blood pressure; pain control. **Note:** Dexmedetomidine causes minimal respiratory depression, inhibits salivation, and is analgesic-sparing.

Critically ill mechanically ventilated ICU patients: Assess and adjust sedation according to scoring system (Richmond Agitation-Sedation Scale [RASS] or Sedation-Agitation Scale [SAS]) (SCCM [Devlin 2018]).

Mechanism of Action

Selective α_2 -adrenoceptor agonist with anesthetic and sedative properties thought to be due to activation of G-proteins by α_{2a} -adrenoceptors in the brainstem resulting in inhibition of norepinephrine release; peripheral α_{2b} -adrenoceptors are activated at high doses or with rapid IV administration resulting in vasoconstriction.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

IV loading dose: 5 to 10 minutes.

Intranasal:

Neonates (PMA: Median: 40.1 weeks): Median: 10 minutes (Bua 2018).

Infants: Median: 10 minutes (Yu 2017).

Children ≤ 12 years: Within 10 to 20 minutes (Miller 2018; Yu 2017).

Adults: 45 to 60 minutes (Yuen 2007).

Peak effect:

IV loading dose: 15 to 30 minutes.

IV continuous infusion: 60 minutes (Barr 2013).

Intranasal: 90 to 105 minutes (Yuen 2007).

Duration, post-continuous infusion (dose dependent): 60 to 240 minutes (Ebert 2000; Hall 2000; Ramsay 2004).

Distribution: IV: V_{ss} :

Preterm Neonates (28 to <36 weeks GA): Median: 2.7 L/kg (range: 2.5 to 5.9 L/kg) (Chrysostomou 2014).

Term Neonates (36 to $\leq$ 44 weeks GA): Median: 3.9 L/kg (range: 0.1 to 10.9 L/kg) (Chrysostomou 2014). Neonates $\geq$ 36 weeks undergoing therapeutic hypothermia for hypoxic ischemic encephalopathy may have a larger volume of distribution (McAdams 2020).

Infants and Children <2 years: Median: 3.8 L/kg (range: 1.9 to 4.6 L/kg) (Vilo 2008).

Children 2 to 11 years: Median: 2.2 L/kg (range: 1.3 to 2.8 L/kg) (Vilo 2008).

Adults: $\sim$ 118 L; rapid.

Bioavailability:

Intramuscular: $73\% \pm 11\%$ (range: 54% to 91%) (Dyck 1993).

Intranasal: Variable: Median: 65% (range: 35% to 93%) (Iriola 2011).

Oral: 16% (range: 12% to 20%) (Anttila 2003).

Sublingual: 72%.

Buccal: 82%.

Protein binding: IV: $\sim$ 94%.

Metabolism: Hepatic via N-glucuronidation, N-methylation, and CYP2A6.

Half-life elimination:

Intramuscular: Terminal: 281 ± 177 minutes (range: 57 to 707 minutes) (Dyck 1993).

IV:

Preterm Neonates (28 to <36 weeks GA): Terminal: Median: 7.6 hours (range: 3 to 9.1 hours) (Chrysostomou 2014).

Term Neonates (36 to $\leq$ 44 weeks GA): Terminal: Median: 3.2 hours (range: 1 to 9.4 hours) (Chrysostomou 2014). Neonates $\geq$ 36 weeks undergoing therapeutic hypothermia for hypoxic ischemic encephalopathy may have a longer elimination half-life (McAdams 2020).

Infants and Children <2 years: Terminal: Median: 2.3 hours (range: 1.5 to 3.3 hours) (Vilo 2008).

Children 2 to 11 years: Terminal: Median: 1.6 hours (range: 1.2 to 2.3 hours) (Vilo 2008).

Adults: Distribution: $\sim$ 6 minutes; Terminal: Approximately up to 3 hours (Venn 2002); significantly prolonged in patients with severe hepatic impairment (Cunningham 1999).

Sublingual/Buccal: 2.8 hours.

Time to peak, serum:

Intramuscular: 13 ± 18 minutes (range: 2 to 60 minutes) (Dyck 1993).

Intranasal: Median: 38 minutes (range: 15 to 60 minutes) (Iriola 2011).

Sublingual/Buccal: 2 hours.

Excretion: IV: Urine (95%); feces (4%).

Clearance: IV:

Note: Clearance following cardiac surgery was reduced by 27% in pediatric patients aged 1 week to 14 years (Potts 2009).

Preterm Neonates (28 to <36 weeks GA): Median: 0.3 L/hour/kg (0.2 to 0.4 L/hour/kg) (Chrysostomou 2014).

Term Neonates (36 to ≤44 weeks GA): Median: 0.9 L/hour/kg (0.2 to 1.5 L/hour/kg) (Chrysostomou 2014). Neonates ≥36 weeks undergoing therapeutic hypothermia for hypoxic ischemic encephalopathy may have lower clearance (McAdams 2020).

Infants and Children <2 years: Median: 1 L/hour/kg (0.85 to 1.66 L/hour/kg) (Vilo 2008).

Children 2 to 11 years: Median: 1 L/hour/kg (0.56 to 1.35 L/hour/kg) (Vilo 2008).

Adults: ~39 L/hour; Hepatic impairment (Child-Pugh class A, B, or C): Mean clearance values were 74%, 64%, and 53% respectively, of those observed in healthy adults.

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Extracorporeal Life Support/Extracorporeal Membrane Oxygenation (ECLS/ECMO) considerations:

No pediatric-specific pharmacokinetic information is available in the literature. Based on the findings of one prospective pharmacokinetic modeling study (n=8, median age: 0.5 years [range: 0.3 to 181.5 months]), clearance may be increased (Thibault 2022).

Removal by ECLS/ECMO circuit: Animal and lab studies suggest approximately 38% to 66% removal by the oxygenator and adsorption to the tubing (Dallefeld 2020; Nasr 2019; Wang 2025). In vitro data suggested that coadministration of other drugs that adsorb to ECLS/ECMO circuit (eg, midazolam) may alter/decrease the concentrations of dexmedetomidine due to changes in binding within circuit (Michihara 2024; Yamasaki 2024).

Drug-specific considerations:

Protein bound, plasma: 94% (Dallefeld 2020; manufacturer's labeling).

Partition coefficient [logP]: 2.8 (very lipophilic) (Dallefeld 2020; PubChem 2025; Wang 2025).

Hepatic function impairment: Clearance and plasma protein binding are decreased in patients with hepatic impairment.

Patient Counseling Points

(For additional information see "Dexmedetomidine: Patient drug information")

What is this drug used for?

Film:

- It is used to treat agitation.

Injection:

- It is used to help you relax during some procedures.
- It is used to cause sleep while on a ventilator.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

Film:

- Feeling sleepy
- Upset stomach
- Numbness or tingling in the mouth
- Dizziness
- Dry mouth

Injection:

- Upset stomach

For a procedure:

- Dry mouth

For use while on a ventilator:

- Constipation

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

- Fast, slow, or abnormal heartbeat
- Chest pain or pressure
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Film:

- Very bad dizziness or passing out

Injection:

- High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Fever
- Trouble breathing, slow breathing, or shallow breathing
- Fast breathing
- Confusion, constipation, diarrhea, dizziness, feeling agitated or nervous, headache, salt cravings, stomach pain, sweating, weakness, or weight loss

For use while on a ventilator:

- Low potassium levels like muscle pain or weakness, muscle cramps, or a heartbeat that does not feel normal
- Feeling agitated
- Feeling nervous and excitable
- Swelling in the arms or legs
- High or low blood sugar like breath that smells like fruit, dizziness, fast breathing, fast heartbeat, feeling confused, feeling sleepy, feeling weak, flushing, headache, unusual thirst or hunger, passing urine more often, shaking, or sweating

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits

of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Dexmedetomidina | Dexmedetomidina gray | Dexmedetomidina richet | Dexmenovag | Dexmetin | Precedex | Valertropina;
- (AT) Austria: Dexdor | Dexmedetomidin ever pharma | Dexmedetomidin mylan | Dexmedetomidin ratiopharm | Dexmedetomidine Accord | Dexmedetomidine kalceks;
- (AU) Australia: Dexdor | Dexmedetomidine Accord | Dexmedetomidine ever pharma | Dexmedetomidine medsurge | Dexmedetomidine mylan | Dexmedetomidine sandoz | Dexmedetomidine teva | Precedex;
- (BE) Belgium: Dexdor | Dexmedetomidine Accord | Dexmedetomidine b. braun | Dexmedetomidine Kabi;
- (BG) Bulgaria: Dexdor | Dexmedetomidin kabi | Dexmedetomidine ever valinject | Dexmedetomidine kalceks;
- (BR) Brazil: Bdexbraun | Cloridrato de dexmedetomidina | Defrik | Dex | Dex bolsa | Extodin | Precedex | Simbilex;
- (CH) Switzerland: Dexdor | Dexmedetomidin labatec | Primadex;
- (CL) Chile: Dexmedetomidina | Precedex;
- (CN) China: You bi tuo;
- (CO) Colombia: Dexdor | Dexmedetomidina | Dexmetinova | Imapren | Kmetodin | Precedex | Redexa | Redexa rtu;
- (CR) Costa Rica: Dexmedetomidina | Precedex | Valertropina;
- (CZ) Czech Republic: Dexdor | Dexmedetomidine Accord | Dexmedetomidine b. braun | Dexmedetomidine ever pharma | Dexmedetomidine Kabi | Dexmedetomidine kalceks | Dexmedetomidine teva | Precedex;
- (DE) Germany: Dexdor | Dexmedetomid accord | Dexmedetomid altan | Dexmedetomid.ethy | Dexmedetomidin altan | Dexmedetomidin b. braun | Dexmedetomidin ethypharm | Dexmedetomidin ever pharma | Dexmedetomidin ever valinject | Dexmedetomidin mylan | Dexmedetomidin ratiopharm;
- (DO) Dominican Republic: Dexmedetomidina | Precedex;
- (EC) Ecuador: Dexmedetomidina | Extodin | Imapren | Precedex | Priaxim;
- (EE) Estonia: Dexdor | Dexmedetomidine Accord | Dexmedetomidine Kabi | Dexmedetomidine kalceks | Dexmedetomidine teva;
- (EG) Egypt: Dexmedetomidine ever pharma | Medrelaxmidine | Precedex;
- (ES) Spain: Dexdor | Dexmedetomidin accord | Dexmedetomidina altan | Dexmedetomidina b. braun | Dexmedetomidina ever pharma | Dexmedetomidina kalceks;
- (FI) Finland: Dexdor | Dexmedetomidine Accord | Dexmedetomidine kalceks | Dexmedetomidine mylan;
- (FR) France: Dexdor | Dexmedetomidine Accord | Dexmedetomidine altan | Dexmedetomidine ever pharma | Dexmedetomidine Kabi | Dexmedetomidine mylan;
- (GB) United Kingdom: Dexdor | Dexmedetomidin | Dexmedetomidine | Dexmedetomidine Accord;
- (GR) Greece: Dexdor;
- (GT) Guatemala: Dexmedetomidina | Dexmedetomidina quimfa | Dexmedetomidina remus;
- (HK) Hong Kong: Dexmedetomidine | Dexmedetomidine orion | Precedex;
- (HU) Hungary: Dexdor | Dexmedetomidin hcl teva | Dexmedetomidine ever pharma | Dexmedetomidine Kabi | Dexmedetomidine kalceks;
- (ID) Indonesia: Neodex op;
- (IE) Ireland: Dexdor | Dexmedetomidin accord | Dexmedetomidine Kabi;
- (IN) India: Alphadex | Dexdine | Dexelex | Dexem | Dexglan | Dexit | Dexmedine | Dextomid |

Sedeto;

- (IT) Italy: Dexdor | Dexmedetomidin ever pharma | Dexmedetomidin mylan | Dexmedetomidina altan | Dexmedetomidina b. braun | Dexmedetomidina teva;
- (JO) Jordan: Precedex;
- (JP) Japan: Dexmedetomidine | Dexmedetomidine sandoz | Precedex;
- (KE) Kenya: Dexem;
- (KR) Korea, Republic of: Dexdine | Dexmedine | Dextomin | Medex | Omnidex | Pamtomidine | Penmix dexmedetomidine hydrochloride | Precedex | Precedex premix | Tomidine;
- (KW) Kuwait: Precedex;
- (LB) Lebanon: Precedex;
- (LT) Lithuania: Dexdor | Dexmedetomidine Accord | Dexmedetomidine kalceks | Dexmedetomidine teva | Dextomid;
- (LV) Latvia: Dexdor | Dexmedetomidine | Dexmedetomidine Accord;
- (MX) Mexico: Avemedit | Danmidrat | Demesynt | Demibindo | Dexmedetomidine | Dexmedetomidine | Dexperum | Drimitec | Kamedix | Krunamina | Meproxiclina | Precedex | Shivaya;
- (MY) Malaysia: Dexmedetomidine Kabi | Emidex;
- (NL) Netherlands: Dexdor | Dexmedetomidin accord | Dexmedetomidine | Dexmedetomidine Kabi | Dexmedetomidine kalceks | Dexmedetomidine teva | Precedex;
- (NO) Norway: Dexdor | Dexmedetomidine b. braun | Dexmedetomidine ever pharma | Dexmedetomidine kalceks | Precedex;
- (NZ) New Zealand: Dexmedetomidine teva | Precedex;
- (PA) Panama: Dexmedetomidina | Dexmedetomidina altan | Dexmedetomidina ever pharma | Dexmedetomidina hidrocloruro | Dexmedetomidina kalceks | Dexmedetomidina richet | Dexmetin | Precedex | Valertropina;
- (PE) Peru: Dexmedetomidina | Dexmepharm | Dexodine | Edetoxin | Precedex;
- (PH) Philippines: Precedex;
- (PL) Poland: Dexdor | Dexmedetomidine ever pharma | Dexmedetomidine kalceks | Dexmedetomidine mylan | Precedex;
- (PR) Puerto Rico: Dexmedetomidin hcl | Dexmedetomidine | Dexmedetomidine HCL | Precedex;
- (PT) Portugal: Dexdor | Dexmedetomidina b. braun | Dexmedetomidina kalceks | Dexmedetomidina mylan | Dexmedetomidina teva | Dexmedetomidine Accord | Precedex;
- (PY) Paraguay: Dexmedetomidina promepar | Dexmedetomidina prosalud | Dexmedotin;
- (QA) Qatar: Precedex | Precedex Ready-to-Use | Sedamid | Semotidin;
- (RO) Romania: Dexdor | Dexmedetomidin ever valinject | Dexmedetomidina kalceks;
- (RU) Russian Federation: Demetodin | Dexdor | Dexmedetomidin | Dexmedetomidine | Dexmedetomidine ever pharma | Dexmedetomidine kalcex | Dexmedin | Dexto;
- (SA) Saudi Arabia: Dexmedetomidine | Precedex | Sedalert | Soporidex;
- (SE) Sweden: Dexdor | Dexmedetomidine Accord | Dexmedetomidine b. braun | Dexmedetomidine ever pharma | Dexmedetomidine kalceks | Dexmedetomidine mylan | Dexmedetomidine teva;
- (SG) Singapore: Dexmedetomidine Kabi | Precedex;
- (SI) Slovenia: Deksmetomidin teva | Dexdor;
- (SK) Slovakia: Dexdor | Dexmedetomidine ever pharma | Dexmedetomidine kalceks | Dexmedetomidine mylan;
- (SV) El Salvador: Precedex;
- (TH) Thailand: Precedex;
- (TN) Tunisia: Dexdor | Precedex;
- (TR) Turkey: Dekstomid | Hipnodex | Precedex | Prexodin | Sedadomid | Semotidin;
- (TW) Taiwan: Dexmedetomidine | Precedex;
- (UA) Ukraine: Dexanest | Dexdor | Dexmedetomidin | Dexmedetomidine ever pharma | Dexmedetomidine novo | Mirodex;
- (UG) Uganda: Dexem;
- (UY) Uruguay: Dexmedetomidina | Dexmedotin | Precedex | Priaxim;
- (VE) Venezuela, Bolivarian Republic of: Sedacons;
- (ZA) South Africa: Demodinex | Dexisun | Eusedex | Sedalpha

Use of UpToDate is subject to the Terms of Use.

REFERENCES

1. Ahmed SS, Unland T, Slaven JE, Nitu ME, Rigby MR. Successful use of intravenous dexmedetomidine for magnetic resonance imaging sedation in autistic children. *South Med J*. 2014;107(9):559-564. [PubMed 25188619]
2. Akin A, Bayram A, Esmoğlu A, et al. Dexmedetomidine vs midazolam for premedication of pediatric patients undergoing anesthesia. *Paediatr Anaesth*. 2012;22(9):871-876. [PubMed 22268591]
3. Alkaissi HR, Khudyakov A, Belligund P. Acute colonic pseudo-obstruction following the use of dexmedetomidine. *Cureus*. 2021;13(11):e19465. doi:10.7759/cureus.19465 [PubMed 34912607]
4. Ala-Kokko TI, Pienimäki P, Lampela E, Hollmén AI, Pelkonen O, Vähäkangas K. Transfer of clonidine and dexmedetomidine across the isolated perfused human placenta. *Acta Anaesthesiol Scand*. 1997;41(2):313-319. doi:10.1111/j.1399-6576.1997.tb04685.x [PubMed 9062619]
5. American Society of Health-System Pharmacists (ASHP). Pediatric continuous infusion standards. <https://www.ashp.org/-/media/assets/pharmacy-practice/s4s/docs/Pediatric-Infusion-Standards.ashx>. Updated March 2024. Accessed March 18, 2024.
6. Anderson PO, Sauberman JB. Modeling drug passage into human milk. *Clin Pharmacol Ther*. 2016;100(1):42-52. doi:10.1002/cpt.377 [PubMed 27060684]
7. Andreolio C, Piva JP, Baldasso E, Ferlini R, Piccoli R. Prolonged infusion of dexmedetomidine in critically-ill children. *Indian Pediatr*. 2016;53(11):987-989. doi:10.1007/s13312-016-0973-2 [PubMed 27889726]
8. Anttila M, Penttilä J, Helminen A, Vuorilehto L, Scheinin H. Bioavailability of dexmedetomidine after extravascular doses in healthy subjects. *Br J Clin Pharmacol*. 2003;56(6):691-693. doi:10.1046/j.1365-2125.2003.01944.x [PubMed 14616431]
9. Aryan HE, Box KW, Ibrahim D, Desiraju U, Ames CP. Safety and efficacy of dexmedetomidine in neurosurgical patients. *Brain Inj*. 2006;20(8):791-798. doi:10.1080/02699050600789447 [PubMed 17060146]
10. Bailey CR. Dexmedetomidine in children - when should we be using it? *Anaesthesia*. 2021;76(3):309-311. doi:10.1111/anae.15169 [PubMed 32578205]
11. Bakan M, Umutoglu T, Topuz U, et al. Opioid-free total intravenous anesthesia with propofol, dexmedetomidine and lidocaine infusions for laparoscopic cholecystectomy: a prospective, randomized, double-blinded study. *Braz J Anesthesiol*. 2015;65(3):191-199. doi:10.1016/j.bjane.2014.05.001 [PubMed 25925031]
12. Bao Z, Zhou C, Wang X, Zhu Y. Intravenous dexmedetomidine during spinal anaesthesia for caesarean section: a meta-analysis of randomized trials. *J Int Med Res*. 2017;45(3):924-932. doi:10.1177/0300060517708945 [PubMed 28553766]
13. Barr J, Fraser GL, Puntillo K, et al; American College of Critical Care Medicine. Clinical practice guidelines for the management of pain, agitation, and delirium in adult patients in the intensive care unit. *Crit Care Med*. 2013;41(1):263-306. doi:10.1097/CCM.0b013e3182783b72 [PubMed 23269131]
14. Based on expert opinion.
15. Bejian S, Valasek C, Nigro JJ, Cleveland DC, Willis BC. Prolonged use of dexmedetomidine in the paediatric cardiothoracic intensive care unit. *Cardiol Young*. 2009;19(1):98-104. [PubMed 19154626]
16. Berkenbosch JW, Wankum PC, Tobias JD. Prospective evaluation of dexmedetomidine for non-invasive procedural sedation in children. *Pediatr Crit Care Med*. 2005;6(4):435-440. [PubMed 15982430]

17. Bharati S, Pal A, Biswas C, Biswas R. Incidence of cardiac arrest increases with the indiscriminate use of dexmedetomidine: a case series and review of published case reports. *Acta Anaesthesiol Taiwan*. 2011;49(4):165-167. doi:10.1016/j.aat.2011.11.010 [PubMed 22221692]
18. Bouajram RH, Bhatt K, Croci R, et al. Incidence of dexmedetomidine withdrawal in adult critically ill patients: a pilot study. *Crit Care Explor*. 2019;1(8):e0035. doi:10.1097/CCE.0000000000000035 [PubMed 32166276]
19. Bromfalk Å, Myrberg T, Walldén J, Engström Å, Hultin M. Preoperative anxiety in preschool children: A randomized clinical trial comparing midazolam, clonidine, and dexmedetomidine. *Paediatr Anaesth*. 2021;31(11):1225-1233. doi:10.1111/pan.14279 [PubMed 34403548]
20. Bua J, Massaro M, Cossovel F, et al. Intranasal dexmedetomidine, as midazolam-sparing drug, for MRI in preterm neonates. *Paediatr Anaesth*. 2018;28(8):747-748. doi:10.1111/pan.13454 [PubMed 30144232]
21. Buck ML. Pharmacokinetic changes during extracorporeal membrane oxygenation: implications for drug therapy of neonates. *Clin Pharmacokinet*. 2003;42(5):403-417. doi:10.2165/00003088-200342050-00001 [PubMed 12739981]
22. Cao Q, Lin Y, Xie Z, et al. Comparison of sedation by intranasal dexmedetomidine and oral chloral hydrate for pediatric ophthalmic examination. *Paediatr Anaesth*. 2017;27(6):629-636. [PubMed 28414899]
23. Carlone G, Trombetta A, Amoroso S, Poropat F, Barbi E, Cozzi G. Intramuscular dexmedetomidine, a feasible option for children with autism spectrum disorders needing urgent procedural sedation. *Pediatr Emerg Care*. 2019;35(6):e116-e117. doi:10.1097/PEC.0000000000001776 [PubMed 31157751]
24. Carroll CL, Krieger D, Campbell M, Fisher DG, Comeau LL, Zucker AR. Use of dexmedetomidine for sedation of children hospitalized in the intensive care unit. *J Hosp Med*. 2008;3(2):142-147. [PubMed 18438790]
25. Cheng V, Abdul-Aziz MH, Roberts JA, Shekar K. Optimising drug dosing in patients receiving extracorporeal membrane oxygenation. *J Thorac Dis*. 2018;10(suppl 5):S629-S641. doi:10.21037/jtd.2017.09.154 [PubMed 29732181]
26. Chrysostomou C, Di Filippo S, Manrique AM, et al. Use of dexmedetomidine in children after cardiac and thoracic surgery. *Pediatr Crit Care Med*. 2006;7(2):126-131. doi:10.1097/01.PCC.0000200967.76996.07 [PubMed 16446599]
27. Chrysostomou C, Sanchez De Toledo J, Avolio T, et al. Dexmedetomidine use in a pediatric cardiac intensive care unit: can we use it in infants after cardiac surgery? *Pediatr Crit Care Med*. 2009;10(6):654-660. doi:10.1097/PCC.0b013e3181a00b7a [PubMed 19295456]
28. Chrysostomou C, Sanchez-de-Toledo J, Wearden P, et al. Perioperative use of dexmedetomidine is associated with decreased incidence of ventricular and supraventricular tachyarrhythmias after congenital cardiac operations. *Ann Thorac Surg*. 2011;92(3):964-72. [PubMed 21871284]
29. Chrysostomou C, Schulman SR, Herrera Castellanos M, et al. A phase II/III, multicenter, safety, efficacy, and pharmacokinetic study of dexmedetomidine in preterm and term neonates. *J Pediatr*. 2014;164(2):276-282. [PubMed 24238862]
30. Cimen ZS, Hanci A, Sivrikaya GU, Kilinc LT, Erol MK. Comparison of buccal and nasal dexmedetomidine premedication for pediatric patients. *Paediatr Anaesth*. 2013;23(2):134-138. [PubMed 22985207]
31. Cortínez LI, Anderson BJ, Holford NH, et al. Dexmedetomidine pharmacokinetics in the obese. *Eur J Clin Pharmacol*. 2015;71(12):1501-1508. doi:10.1007/s00228-015-1948-2 [PubMed 26407689]
32. Cosnahan AS, Angert RM, Jano E, Wachtel EV. Dexmedetomidine versus intermittent morphine for sedation of neonates with encephalopathy undergoing therapeutic hypothermia. *J Perinatol*. 2021;41(9):2284-2291. doi:10.1038/s41372-021-00998-8 [PubMed 33649447]

33. Cree ML, Abdul-Aziz MH, Schlapbach LJ, Roberts JA, Parker SL. The impact of extracorporeal support on antimicrobial pharmacokinetics in critically ill neonatal and paediatric patients: a systematic review. *Int J Antimicrob Agents*. 2024;64(4):107311. doi:10.1016/j.ijantimicag.2024.107311 [PubMed 39197687]
34. Cunningham FE, Baughman VL, Tonkovich L, et al. Pharmacokinetics of dexmedetomidine (DEX) in patients with hepatic failure (HF). *Clin Pharmacol Ther*. 1999;65:128.
35. Czaja AS, Zimmerman JJ. The use of dexmedetomidine in critically ill children. *Pediatr Crit Care Med*. 2009;10(3):381-386. [PubMed 19325505]
36. Dallefeld SH, Sherwin J, Zimmerman KO, Watt KM. Dexmedetomidine extraction by the extracorporeal membrane oxygenation circuit: results from an in vitro study. *Perfusion*. 2020;35(3):209-216. doi:10.1177/0267659119868062 [PubMed 31431126]
37. Darnell C, Steiner J, Szmuk P, Sheeran P. Withdrawal from multiple sedative agent therapy in an infant: is dexmedetomidine the cause or the cure? *Pediatr Crit Care Med*. 2010;11(1):e1-3. doi:10.1097/PCC.0b013e3181a66131 [PubMed 20051785]
38. Davy A, Fessler J, Fischler M, LE Guen M. Dexmedetomidine and general anesthesia: a narrative literature review of its major indications for use in adults undergoing non-cardiac surgery. *Minerva Anesthesiol*. 2017;83(12):1294-1308. doi:10.23736/S0375-9393.17.12040-7 [PubMed 28643999]
39. De Baerdemaeker L, Margaron M. Best anaesthetic drug strategy for morbidly obese patients. *Curr Opin Anaesthesiol*. 2016;29(1):119-128. doi:10.1097/ACO.0000000000000286 [PubMed 26658181]
40. De Wolf AM, Fragen RJ, Avram MJ, Fitzgerald PC, Rahimi-Danesh F. The pharmacokinetics of dexmedetomidine in volunteers with severe renal impairment. *Anesth Analg*. 2001;93(5):1205-1209. doi:10.1097/00000539-200111000-00031. [PubMed 11682398]
41. Dersch-Mills DA, Banasch HL, Yusuf K, Howlett A. Dexmedetomidine use in a tertiary care NICU: a descriptive study. *Ann Pharmacother*. 2019;53(5):464-470. doi:10.1177/1060028018812089 [PubMed 30501499]
42. Devlin JW, Skrobik Y, Gélinas C, et al. Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med*. 2018;46(9):e825-e873. doi:10.1097/CCM.0000000000003299. [PubMed 30113379]
43. Dexmedetomidine multiple-dose vials and premixed bags in D5W Injection [prescribing information]. Paramus, NJ; WG Critical Care LLC: May 2026.
44. Dodd SE, Hunter Guevara LR, Datta P, Rewers-Felkins K, Baker T, Hale TW. Dexmedetomidine levels in breast milk: analysis of breast milk expressed during and after awake craniotomy. *Breastfeed Med*. 2021;16(11):919-921. doi:10.1089/bfm.2021.0138 [PubMed 34143658]
45. Dutta A, Sethi N, Sood J, et al. The effect of dexmedetomidine on propofol requirements during anesthesia administered by bispectral index-guided closed-loop anesthesia delivery system: a randomized controlled study. *Anesth Analg*. 2019;129(1):84-91. doi:10.1213/ANE.0000000000003470 [PubMed 29787410]
46. Dyck JB, Maze M, Haack C, Vuorilehto L, Shafer SL. The pharmacokinetics and hemodynamic effects of intravenous and intramuscular dexmedetomidine hydrochloride in adult human volunteers. *Anesthesiology*. 1993;78(5):813-820. doi:10.1097/00000542-199305000-00002 [PubMed 8098190]
47. Ebert TJ, Hall JE, Barney JA, Uhrich TD, Colino MD. The effects of increasing plasma concentrations of dexmedetomidine in humans. *Anesthesiology*. 2000;93(2):382-394. doi:10.1097/00000542-200008000-00016 [PubMed 10910487]

48. El Amrousy DM, Elshmaa NS, El-Kashlan M, et al. Efficacy of prophylactic dexmedetomidine in preventing postoperative junctional ectopic tachycardia after pediatric cardiac surgery. *J Am Heart Assoc.* 2017;6(3):e004780. doi:10.1161/JAHA.116.004780 [PubMed 28249845]
49. Elliott M, Burnsed J, Heinan K, et al. Effect of dexmedetomidine on heart rate in neonates with hypoxic ischemic encephalopathy undergoing therapeutic hypothermia. *J Neonatal Perinatal Med.* 2022;15(1):47-54. doi:10.3233/NPM-210737 [PubMed 34334427]
50. Enomoto Y, Kudo T, Saito T, et al. Prolonged use of dexmedetomidine in an infant with respiratory failure following living donor liver transplantation. *Paediatr Anaesth.* 2006;16(12):1285-1288. doi:10.1111/j.1460-9592.2006.02008.x [PubMed 17121562]
51. Erdman MJ, Doepker BA, Gerlach AT, Phillips GS, Elijevich L, Jones GM. A comparison of severe hemodynamic disturbances between dexmedetomidine and propofol for sedation in neurocritical care patients. *Crit Care Med.* 2014;42(7):1696-1702. doi:10.1097/CCM.0000000000000328 [PubMed 24717468]
52. Erstad BL, Barletta JF. Drug dosing in the critically ill obese patient-a focus on sedation, analgesia, and delirium. *Crit Care.* 2020;24(1):315. doi:10.1186/s13054-020-03040-z [PubMed 32513237]
53. Eskandr AM, Metwally AA, Ahmed AA, et al. Dexmedetomidine as a part of general anaesthesia for caesarean delivery in patients with pre-eclampsia: a randomised double-blinded trial. *Eur J Anaesthesiol.* 2018;35(5):372-378. doi:10.1097/EJA.0000000000000776 [PubMed 29432379]
54. Estkowski LM, Morris JL, Sinclair EA. Characterization of dexmedetomidine dosing and safety in neonates and infants. *J Pediatr Pharmacol Ther.* 2015;20(2):112-118. doi:10.5863/1551-6776-20.2.112 [PubMed 25964728]
55. Expert opinion. Senior Hepatic Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCPS, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.
56. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.
57. Flieller LA, Alaniz C, Pleva MR, Miller JT. Incidence of rebound hypertension after discontinuation of dexmedetomidine. *Pharmacotherapy.* 2019;39(10):970-974. doi:10.1002/phar.2323 [PubMed 31463963]
58. Franciscovich CD, Monk HM, Brodecki D, et al. Sedation practices of neonates receiving extracorporeal membrane oxygenation. *ASAIO J.* 2020;66(5):559-564. doi:10.1097/MAT.0000000000001043 [PubMed 31425254]
59. Gerlach AT, Blais DM, Jones GM, et al. Predictors of dexmedetomidine-associated hypotension in critically ill patients. *Int J Crit Illn Inj Sci.* 2016;6(3):109-114. doi:10.4103/2229-5151.190656 [PubMed 27722111]
60. Gerlach AT, Dasta JF, Steinberg S, Martin LC, Cook CH. A new dosing protocol reduces dexmedetomidine-associated hypotension in critically ill surgical patients. *J Crit Care.* 2009;24(4):568-574. doi:10.1016/j.jcrc.2009.05.015 [PubMed 19682844]
61. Gerlach AT, Murphy CV. Sedation with dexmedetomidine in the intensive care setting. *Open Access Emerg Med.* 2011;3:77-85. doi:10.2147/OAEM.S17429 [PubMed 27147855]
62. Ghai B, Jain K, Saxena AK, Bhatia N, Sodhi KS. Comparison of oral midazolam with intranasal dexmedetomidine premedication for children undergoing CT imaging: a randomized, double-blind, and controlled study. *Paediatr Anaesth.* 2017;27(1):37-44. [PubMed 27734549]
63. Goettel N, Bharadwaj S, Venkatraghavan L, Mehta J, Bernstein M, Manninen PH. Dexmedetomidine vs propofol-remifentanyl conscious sedation for awake craniotomy: a prospective randomized controlled trial. *Br J Anaesth.* 2016;116(6):811-821. doi:10.1093/bja/aew024 [PubMed 27099154]
64. Gong M, Man Y, Fu Q. Incidence of bradycardia in pediatric patients receiving dexmedetomidine

- anesthesia: a meta-analysis. *Int J Clin Pharm*. 2017;39(1):139-147. doi:10.1007/s11096-016-0411-5 [PubMed 28040841]
65. Greenberg RG, Wu H, Laughon M, et al. Population pharmacokinetics of dexmedetomidine in infants. *J Clin Pharmacol*. 2017;57(9):1174-1182. [PubMed 28444697]
 66. Guinter JR, Kristeller JL. Prolonged infusions of dexmedetomidine in critically ill patients. *Am J Health Syst Pharm*. 2010;67(15):1246-1253. doi:10.2146/ajhp090300 [PubMed 20651314]
 67. Gyanesh P, Haldar R, Srivastava D, Agrawal PM, Tiwari AK, Singh PK. Comparison between intranasal dexmedetomidine and intranasal ketamine as premedication for procedural sedation in children undergoing MRI: a double-blind, randomized, placebo-controlled trial. *J Anesth*. 2014;28(1):12-18. [PubMed 23800984]
 68. Haenecour AS, Seto W, Urbain CM, Stephens D, Laussen PC, Balit CR. Prolonged dexmedetomidine infusion and drug withdrawal In critically ill children. *J Pediatr Pharmacol Ther*. 2017;22(6):453-460. doi:10.5863/1551-6776-22.6.453 [PubMed 29290746]
 69. Hall JE, Uhrich TD, Barney JA, Arain SR, Ebert TJ. Sedative, amnestic, and analgesic properties of small-dose dexmedetomidine infusions. *Anesth Analg*. 2000;90(3):699-705. doi:10.1097/00000539-200003000-00035 [PubMed 10702460]
 70. Hammer GB, Drover DR, Cao H, et al. The effects of dexmedetomidine on cardiac electrophysiology in children. *Anesth Analg*. 2008;106(1):79-83, table of contents. doi:10.1213/01.ane.0000297421.9285 [PubMed 18165557]
 71. Heard C, Burrows F, Johnson K, Joshi P, Houck J, Lerman J. A comparison of dexmedetomidine-midazolam with propofol for maintenance of anesthesia in children undergoing magnetic resonance imaging. *Anesth Analg*. 2008;107(6):1832-1839. doi:10.1213/ane.0b013e31818874ee [PubMed 19020127]
 72. Hosokawa K, Shime N, Kato Y, et al. Dexmedetomidine sedation in children after cardiac surgery. *Pediatr Crit Care Med*. 2010;11(1):39-43. [PubMed 19593247]
 73. Hughes CG, Mailloux PT, Devlin JW, et al; MENDS2 Study Investigators. Dexmedetomidine or propofol for sedation in mechanically ventilated adults with sepsis. *N Engl J Med*. 2021;384(15):1424-1436. doi:10.1056/NEJMoa2024922 [PubMed 33528922]
 74. Ibrahim M. A prospective, randomized, double blinded comparison of intranasal dexmedetomidine vs intranasal ketamine in combination with intravenous midazolam for procedural sedation in school aged children undergoing MRI. *Anesth Essays Res*. 2014;8(2):179-186. [PubMed 25886223]
 75. Ice CJ, Personett HA, Frazee EN, Dierkhising RA, Kashyap R, Oeckler RA. Risk factors for dexmedetomidine-associated hemodynamic instability in noncardiac intensive care unit patients. *Anesth Analg*. 2016;122(2):462-69. doi:10.1213/ANE.0000000000001125 [PubMed 26702868]
 76. Igalmi (dexmedetomidine) [prescribing information]. New Haven, CT: BioXcel Therapeutics Inc; November 2025.
 77. Iriola T, Vilo S, Manner T, et al. Bioavailability of dexmedetomidine after intranasal administration. *Eur J Clin Pharmacol*. 2011;67(8):825-831. [PubMed 21318594]
 78. Ingersoll-Weng E, Manecke GR Jr, Thistlethwaite PA. Dexmedetomidine and cardiac arrest. *Anesthesiology*. 2004;100(3):738-739. doi:10.1097/00000542-200403000-00040 [PubMed 15108994]
 79. Institute for Safe Medication Practices (ISMP). Do you know what doses are being programmed in the OR? Make it an expectation to use smart infusion pumps with DERS. <https://www.ismp.org/resources/do-you-know-what-doses-are-being-programmed-or-make-it-expectation-use-smart-infusion>. Published March 12, 2020. Accessed June 19, 2020.

80. Institute for Safe Medication Practices (ISMP). Drug shortages continue to compromise patient care. <https://www.ismp.org/resources/drug-shortages-continue-compromise-patient-care>. Published January 11, 2018. Accessed June 19, 2020.
81. Ito S. Drug therapy for breast-feeding women. *NEJM*. 2000;343(2):118-126. [PubMed 10891521]
82. Jakob SM, Ruokonen E, Grounds RM, et al; Dexmedetomidine for Long-Term Sedation Investigators. Dexmedetomidine vs midazolam or propofol for sedation during prolonged mechanical ventilation: two randomized controlled trials. *JAMA*. 2012;307(11):1151-1160. doi:10.1001/jama.2012.304 [PubMed 22436955]
83. Janmahasatian S, Duffull SB, Ash S, Ward LC, Byrne NM, Green B. Quantification of lean bodyweight. *Clin Pharmacokinet*. 2005;44(10):1051-1065. doi:10.2165/00003088-200544100-00004 [PubMed 16176118]
84. Ji F, Liu H. Intraoperative hypernatremia and polyuric syndrome induced by dexmedetomidine. *J Anesth*. 2013;27(4):599-603. doi:10.1007/s00540-013-1562-3 [PubMed 23377505]
85. Joo EY, Kim YJ, Park YS, et al. Intramuscular dexmedetomidine and oral chloral hydrate for pediatric sedation for electroencephalography: a propensity score-matched analysis. *Paediatr Anaesth*. 2020;30(5):584-591. doi:10.1111/pan.13844 [PubMed 32107813]
86. Jooste EH, Muhly WT, Ibinson JW, et al. Acute hemodynamic changes after rapid intravenous bolus dosing of dexmedetomidine in pediatric heart transplant patients undergoing routine cardiac catheterization. *Anesth Analg*. 2010;111(6):1490-1496. doi:10.1213/ANE.0b013e3181f7e2ab [PubMed 21059743]
87. Kadam SV, Tailor KB, Kulkarni S, Mohanty SR, Joshi PV, Rao SG. Effect of dexmedetomidine on postoperative junctional ectopic tachycardia after complete surgical repair of tetralogy of Fallot: A prospective randomized controlled study. *Ann Card Anaesth*. 2015;18(3):323-328. doi:10.4103/0971-9784.159801 [PubMed 26139736]
88. Kaur M, Singh PM. Current role of dexmedetomidine in clinical anesthesia and intensive care. *Anesth Essays Res*. 2011;5(2):128-133. doi:10.4103/0259-1162.94750 [PubMed 25885374]
89. Keles S, Kocaturk O. Comparison of oral dexmedetomidine and midazolam for premedication and emergence delirium in children after dental procedures under general anesthesia: a retrospective study. *Drug Des Devel Ther*. 2018;12:647-653. doi:10.2147/DDDT.S163828 [PubMed 29636599]
90. Keles S, Kocaturk O. The effect of oral dexmedetomidine premedication on preoperative cooperation and emergence delirium in children undergoing dental procedures. *Biomed Res Int*. 2017;2017:6742183. doi:10.1155/2017/6742183 [PubMed 28904966]
91. Kim BJ, Kim BI, Byun SH, Kim E, Sung SY, Jung JY. Cardiac arrest in a patient with anterior fascicular block after administration of dexmedetomidine with spinal anesthesia: A case report. *Medicine (Baltimore)*. 2016;95(43):e5278. doi:10.1097/MD.0000000000005278 [PubMed 27787391]
92. Kim JY, Kim KN, Kim DW, Lim HJ, Lee BS. Effects of dexmedetomidine sedation for magnetic resonance imaging in children: a systematic review and meta-analysis. *J Anesth*. 2021;35(4):525-535. doi:10.1007/s00540-021-02946-4 [PubMed 34002258]
93. Koroglu A, Teksan H, Sagir O, Yucel A, Toprak HI, Ersoy OM. A comparison of the sedative, hemodynamic, and respiratory effects of dexmedetomidine and propofol in children undergoing magnetic resonance imaging. *Anesth Analg*. 2006;103(1):63-67. [PubMed 16790627]
94. Le Guen M, Liu N, Tounou F, et al. Dexmedetomidine reduces propofol and remifentanyl requirements during bispectral index-guided closed-loop anesthesia: a double-blind, placebo-controlled trial. *Anesth Analg*. 2014;118(5):946-955. doi:10.1213/ANE.0000000000000185 [PubMed 24722260]
95. Lee M, Kim H, Lee C, Kang H. Effect of intravenous dexmedetomidine and remifentanyl on neonatal outcomes after caesarean section under general anaesthesia: a systematic review and meta-

- analysis. *Eur J Anaesthesiol*. 2021;38(10):1085-1095. doi:10.1097/EJA.0000000000001558 [PubMed 34101715]
96. Lewis K, Balas MC, Stollings JL, et al. A focused update to the clinical practice guidelines for the prevention and management of pain, anxiety, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med*. 2025;53(3):e711-e727. doi:10.1097/CCM.0000000000006574 [PubMed 39982143]
97. Li BL, Yuen VM, Song XR, et al. Intranasal dexmedetomidine following failed chloral hydrate sedation in children. *Anaesthesia*. 2014;69(3):240-244. doi:10.1111/anae.12533 [PubMed 24447296]
98. Li L, Zhou J, Yu D, Hao X, Xie Y, Zhu T. Intranasal dexmedetomidine versus oral chloral hydrate for diagnostic procedures sedation in infants and toddlers: A systematic review and meta-analysis. *Medicine (Baltimore)*. 2020;99(9):e19001. [PubMed 32118711]
99. Li X, Zhang C, Dai D, Liu H, Ge S. Efficacy of dexmedetomidine in prevention of junctional ectopic tachycardia and acute kidney injury after pediatric cardiac surgery: A meta-analysis. *Congenit Heart Dis*. 2018;13(5):799-807. doi:10.1111/chd.12674 [PubMed 30260073]
100. Liu S, Peng P, Hu Y, et al. The effectiveness and safety of intravenous dexmedetomidine of different concentrations combined with butorphanol for post-caesarean section analgesia: a randomized controlled trial. *Drug Des Devel Ther*. 2021;15:689-698. doi:10.2147/DDDT.S287512 [PubMed 33628014]
101. Liu X, Zhang X, Wang X, Wang J, Wang H. Comparative evaluation of intrathecal bupivacaine alone and bupivacaine combined with dexmedetomidine in cesarean section using spinal anesthesia: a meta-analysis. *J Int Med Res*. 2019;47(7):2785-2799. doi:10.1177/0300060518797000 [PubMed 31204535]
102. Mahmoud M, Mason KP. Dexmedetomidine: review, update, and future considerations of paediatric perioperative and periprocedural applications and limitations. *Br J Anaesth*. 2015;115(2):171-182. doi:10.1093/bja/aeu226 [PubMed 26170346]
103. Mateos Gaitan R, Vicent L, Rodriguez-Queralto O, et al. Dexmedetomidine in medical cardiac intensive care units. Data from a multicenter prospective registry. *Int J Cardiol*. 2020;310:162-166. doi:10.1016/j.ijcard.2020.04.002 [PubMed 32307185]
104. Mason KP, Lubisch NB, Robinson F, Roskos R. Intramuscular dexmedetomidine sedation for pediatric MRI and CT. *AJR Am J Roentgenol*. 2011;197(3):720-725. doi:10.2214/AJR.10.6134 [PubMed 21862817]
105. Mason KP, Lubisch N, Robinson F, Roskos R, Epstein MA. Intramuscular dexmedetomidine: an effective route of sedation preserves background activity for pediatric electroencephalograms. *J Pediatr*. 2012;161(5):927-932. doi:10.1016/j.jpeds.2012.05.011 [PubMed 22704249]
106. Mason KP, Park RS, Sullivan CA, et al. The synergistic effect of dexmedetomidine on propofol for paediatric deep sedation: a randomised trial. *Eur J Anaesthesiol*. 2021;38(5):541-547. doi:10.1097/EJA.0000000000001350 [PubMed 33009191]
107. Mason KP, Robinson F, Fontaine P, Prescilla R. Dexmedetomidine offers an option for safe and effective sedation for nuclear medicine imaging in children. *Radiology*. 2013;267(3):911-917. [PubMed 23449958]
108. Mason KP, Zurakowski D, Zgleszewski SE, et al. High dose dexmedetomidine as the sole sedative for pediatric MRI. *Paediatr Anaesth*. 2008;18(5):403-411. doi:10.1111/j.1460-9592.2008.02468.x [PubMed 18363626]
109. McAdams RM, Pak D, Lalovic B, Phillips B, Shen DD. Dexmedetomidine pharmacokinetics in neonates with hypoxic-ischemic encephalopathy receiving hypothermia. *Anesthesiol Res Pract*. 2020;2020:2582965. doi:10.1155/2020/2582965 [PubMed 32158472]
110. Mendel B, Christianto C, Setiawan M, Prakoso R, Siagian SN. A comparative effectiveness systematic review and meta-analysis of drugs for the prophylaxis of junctional ectopic tachycardia.

Curr Cardiol Rev. 2022;18(1):e030621193817. doi:10.2174/1573403X17666210603113430 [PubMed 34082685]

111. Michihara A, Hanada M, Nagatsuka Handa Y, Mizoguchi T, Ohchi Y, Sato Y. Change of dexmedetomidine and midazolam concentrations by simultaneous injection in an in vitro extracorporeal circuit. *Perfusion.* 2024;39(6):1222-1230. doi:10.1177/02676591231184715 [PubMed 37325845]
112. Miller JW, Balyan R, Dong M, et al. Does intranasal dexmedetomidine provide adequate plasma concentrations for sedation in children: a pharmacokinetic study. *Br J Anaesth.* 2018;120(5):1056-1065. doi:10.1016/j.bja.2018.01.035 [PubMed 29661383]
113. Mitchell J, Jones W, Winkley E, Kinsella SM. Dexmedetomidine in breast milk: a reply. *Anaesthesia.* 2021;76(2):289. doi:10.1111/anae.15278 [PubMed 33090472]
114. Mitchell J, Jones W, Winkley E, Kinsella SM. Guideline on anaesthesia and sedation in breastfeeding women 2020: guideline from the Association of Anaesthetists. *Anaesthesia.* 2020;75(11):1482-1493. doi:10.1111/anae.15179 [PubMed 32737881]
115. Morse JD, Cortinez LI, Anderson BJ. A universal pharmacokinetic model for dexmedetomidine in children and adults. *J Clin Med.* 2020;9(11):3480. doi:10.3390/jcm9113480 [PubMed 33126702]
116. Nakanishi R, Yoshimura M, Suno M, et al. Detection of dexmedetomidine in human breast milk using liquid chromatography-tandem mass spectrometry: application to a study of drug safety in breastfeeding after cesarean section. *J Chromatogr B Analyt Technol Biomed Life Sci.* 2017;1040:208-213. doi:10.1016/j.jchromb.2016.11.015 [PubMed 27856195]
117. Nasr VG, Meserve J, Pereira LM, et al. Sedative and analgesic drug sequestration after a single bolus injection in an ex vivo extracorporeal membrane oxygenation infant circuit. *ASAIO J.* 2019;65(2):187-191. doi:10.1097/MAT.0000000000000793 [PubMed 29595531]
118. National Center for Biotechnology Information (NCBI). PubChem: dexmedetomidine. <https://pubchem.ncbi.nlm.nih.gov/compound/Dexmedetomidine>. Accessed November 14, 2025.
119. Naveed M, Bondi DS, Shah PA. Dexmedetomidine versus fentanyl for neonates with hypoxic ischemic encephalopathy undergoing therapeutic hypothermia. *J Pediatr Pharmacol Ther.* 2022;27(4):352-357. doi:10.5863/1551-6776-27.4.352 [PubMed 35558346]
120. Nie Y, Tu W, Shen X, et al. Dexmedetomidine added to sufentanil patient-controlled intravenous analgesia relieves the postoperative pain after cesarean delivery: a prospective randomized controlled multicenter study. *Sci Rep.* 2018;8(1):9952. doi:10.1038/s41598-018-27619-3 [PubMed 29967332]
121. O'Mara K, Gal P, Ransommd JL, et al. Successful use of dexmedetomidine for sedation in a 24-week gestational age neonate. *Ann Pharmacother.* 2009;43(10):1707-1713. [PubMed 19755621]
122. O'Mara K, Gal P, Wimmer J, et al. Dexmedetomidine versus standard therapy with fentanyl for sedation in mechanically ventilated premature neonates. *J Pediatr Pharmacol Ther.* 2012;17(3):252-262. [PubMed 23258968]
123. O'Mara K, Weiss MD. Dexmedetomidine for sedation of neonates with HIE undergoing therapeutic hypothermia: a single-center experience. *AJP Rep.* 2018;8(3):e168-e173. [PubMed 30186671]
124. Page RL 2nd, O'Bryant CL, Cheng D, et al; American Heart Association Clinical Pharmacology and Heart Failure and Transplantation Committees of the Council on Clinical Cardiology; Council on Cardiovascular Surgery and Anesthesia; Council on Cardiovascular and Stroke Nursing; and Council on Quality of Care and Outcomes Research. Drugs That May Cause or Exacerbate Heart Failure: A Scientific Statement From the American Heart Association [published correction appears in *Circulation.* 2016;134(12):e261]. *Circulation.* 2016;134(6):e32-e69. [PubMed 27400984]

125. Patel CR, Engineer SR, Shah BJ, Madhu S. Effect of intravenous infusion of dexmedetomidine on perioperative haemodynamic changes and postoperative recovery: A study with entropy analysis. *Indian J Anaesth.* 2012;56(6):542-546. doi:10.4103/0019-5049.104571 [PubMed 23325938]
126. Patel JS, Kooda K, Ignieri LA. A narrative review of the impact of extracorporeal membrane oxygenation on the pharmacokinetics and pharmacodynamics of critical care therapies. *Ann Pharmacother.* 2023;57(6):706-726. doi:10.1177/10600280221126438 [PubMed 36250355]
127. Pathan S, Kaplan JB, Adamczyk K, Chiu SH, Shah CV. Evaluation of dexmedetomidine withdrawal in critically ill adults. *J Crit Care.* 2021;62:19-24. doi:10.1016/j.jcrc.2020.10.024 [PubMed 33227592]
128. Phelps JR, Russell A, Lupa MC, et al. High-dose dexmedetomidine for noninvasive pediatric procedural sedation and discharge readiness. *Paediatr Anaesth.* 2015;25(9):877-882. doi:10.1111/pan.12569 [PubMed 25565076]
129. Potts AL, Anderson BJ, Warman GR, Lerman J, Diaz SM, Vilo S. Dexmedetomidine pharmacokinetics in pediatric intensive care--a pooled analysis. *Paediatr Anaesth.* 2009;19(11):1119-1129. [PubMed 19708909]
130. Precedex (dexmedetomidine) [prescribing information]. Lake Forest, IL: Hospira Inc; May 2026.
131. Precedex (dexmedetomidine) [product monograph]. Kirkland, Quebec, Canada: Pfizer Canada ULC; August 2024.
132. Preslaski CR, Mueller SW, Wempe MF, MacLaren R. Stability of dexmedetomidine in polyvinyl chloride bags containing 0.9% sodium chloride injection. *Am J Health Syst Pharm.* 2013;70(15):1336-1341. doi:10.2146/ajhp120390 [PubMed 23867490]
133. Ramsay MA, Luterman DL. Dexmedetomidine as a total intravenous anesthetic agent. *Anesthesiology.* 2004;101(3):787-790. doi:10.1097/00000542-200409000-00028 [PubMed 15329604]
134. Refer to manufacturer's labeling.
135. Reynolds J, Rogers A, Capehart S, Manyang P, Watcha MF. Retrospective comparison of intranasal dexmedetomidine and oral chloral hydrate for sedated auditory brainstem response exams. *Hosp Pediatr.* 2016;6(3):166-171. [PubMed 26917547]
136. Riker RR, Shehabi Y, Bokesch PM, et al; SEDCOM (Safety and Efficacy of Dexmedetomidine Compared With Midazolam) Study Group. Dexmedetomidine vs midazolam for sedation of critically ill patients: a randomized trial. *JAMA.* 2009;301(5):489-99. doi:10.1001/jama.2009.56 [PubMed 19188334]
137. Rolle A, Paredes S, Cortínez LI, et al. Dexmedetomidine metabolic clearance is not affected by fat mass in obese patients. *Br J Anaesth.* 2018;120(5):969-977. doi:10.1016/j.bja.2018.01.040 [PubMed 29661414]
138. Sajid B, Mohamed T, Jumaila M. A comparison of oral dexmedetomidine and oral midazolam as premedicants in children. *J Anaesthesiol Clin Pharmacol.* 2019;35(1):36-40. doi:10.4103/joacp.JOACP_20_18 [PubMed 31057237]
139. Salah J, Grgurich P, Nault K, Lei Y. Identification of risk factors for hypertension and tachycardia upon dexmedetomidine discontinuation. *J Crit Care.* 2020;59:81-85. doi:10.1016/j.jcrc.2020.06.006 [PubMed 32580122]
140. Schwartz LI, Miyamoto SD, Stenquist S, Twite MD. Cardiac arrest in a heart transplant patient receiving dexmedetomidine during cardiac catheterization. *Semin Cardiothorac Vasc Anesth.* 2016;20(2):175-178. doi:10.1177/1089253215624765 [PubMed 26721807]
141. Shah AN, Koneru J, Nicoara A, Goldfeder LB, Thomas K, Ehlert FA. Dexmedetomidine related cardiac arrest in a patient with permanent pacemaker; a cautionary tale. *Pacing Clin Electrophysiol.* 2007;30(9):1158-1160. doi:10.1111/j.1540-8159.2007.00829.x [PubMed 17725762]

142. Shehabi Y, Howe BD, Bellomo R, et al; ANZICS Clinical Trials Group and the SPICE III Investigators. Early sedation with dexmedetomidine in critically ill patients. *N Engl J Med*. 2019;380(26):2506-2517. doi:10.1056/NEJMoa1904710 [PubMed 31112380]
143. Shehabi Y, Ruettimann U, Adamson H, Innes R, Ickeringill M. Dexmedetomidine infusion for more than 24 hours in critically ill patients: sedative and cardiovascular effects. *Intensive Care Med*. 2004;30(12):2188-2196. doi:10.1007/s00134-004-2417-z [PubMed 15338124]
144. Shen QH, Li HF, Zhou XY, Yuan XZ, Lu YP. Dexmedetomidine as an adjuvant for single spinal anesthesia in patients undergoing cesarean section: a system review and meta-analysis. *J Int Med Res*. 2020;48(5):300060520913423. doi:10.1177/0300060520913423 [PubMed 32466699]
145. Shen SL, Zheng JY, Zhang J, et al. Comparison of dexmedetomidine and propofol for conscious sedation in awake craniotomy: a prospective, double-blind, randomized, and controlled clinical trial. *Ann Pharmacother*. 2013;47(11):1391-1399. doi:10.1177/1060028013504082 [PubMed 24259599]
146. Shin HJ, Do SH, Lee JS, Kim TK, Na HS. Comparison of intraoperative sedation with dexmedetomidine versus propofol on acute postoperative pain in total knee arthroplasty under spinal anesthesia: a randomized trial. *Anesth Analg*. 2019;129(6):1512-1518. doi:10.1213/ANE.0000000000003315 [PubMed 31743170]
147. Siddappa R, Riggins J, Kariyanna S, Calkins P, Rotta AT. High-dose dexmedetomidine sedation for pediatric MRI. *Paediatr Anaesth*. 2011;21(2):153-158. [PubMed 21210884]
148. Sim JH, Yu HJ, Kim ST. The effects of different loading doses of dexmedetomidine on sedation. *Korean J Anesthesiol*. 2014;67(1):8-12. doi:10.4097/kjae.2014.67.1.8 [PubMed 25097732]
149. Smith HAB, Besunder JB, Betters KA, et al. 2022 Society of Critical Care Medicine clinical practice guidelines on prevention and management of pain, agitation, neuromuscular blockade, and delirium in critically ill pediatric patients with consideration of the ICU environment and early mobility. *Pediatr Crit Care Med*. 2022;23(2):e74-e110. doi:10.1097/PCC.0000000000002873 [PubMed 35119438]
150. Song JC, Gao H, Qiu HB, et al. The pharmacokinetics of dexmedetomidine in patients with obstructive jaundice: a clinical trial. *PLoS One*. 2018;13(11):e0207427. doi:10.1371/journal.pone.0207427 [PubMed 30427948]
151. Stovall J. Bipolar mania and hypomania in adults: choosing pharmacotherapy. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 6, 2022.
152. Su F, Gastonguay MR, Nicolson SC, DiLiberto M, Ocampo-Pelland A, Zuppa AF. Dexmedetomidine pharmacology in neonates and infants after open heart surgery. *Anesth Analg*. 2016;122(5):1556-1566. [PubMed 26218862]
153. Sulton C, Kamat P, Mallory M, Reynolds J. The use of intranasal dexmedetomidine and midazolam for sedated magnetic resonance imaging in children: a report from the Pediatric Sedation Research Consortium. *Pediatr Emerg Care*. 2020;36(3):138-142. [PubMed 28609332]
154. Sun S, Wang J, Wang J, Wang F, Xia H, Yao S. Fetal and maternal responses to dexmedetomidine intrathecal application during cesarean section: a meta-analysis. *Med Sci Monit*. 2020;26:e918523. doi:10.12659/MSM.918523 [PubMed 31995551]
155. Surkov D. Is dexmedetomidine a potential neuroprotective agent for term neonates with hypoxic-ischemic encephalopathy? *PACCJ*. 2019;7(1):22-30. doi:10.14587/paccj.2019.4
156. Takata K, Adachi YU, Suzuki K, Obata Y, Sato S, Nishiwaki K. Dexmedetomidine-induced atrioventricular block followed by cardiac arrest during atrial pacing: a case report and review of the literature. *J Anesth*. 2014;28(1):116-120. doi:10.1007/s00540-013-1676-7 [PubMed 23948748]
157. Talon MD, Woodson LC, Sherwood ER, Aarsland A, McRae L, Benham T. Intranasal dexmedetomidine premedication is comparable with midazolam in burn children undergoing reconstructive surgery. *J Burn Care Res*. 2009;30(4):599-605. [PubMed 19506498]

158. Tan JA, Ho KM. Use of dexmedetomidine as a sedative and analgesic agent in critically ill adult patients: a meta-analysis. *Intensive Care Med.* 2010;36(6):926-939. doi:10.1007/s00134-010-1877-6 [PubMed 20376429]
159. Tellor BR, Arnold HM, Micek ST, Kollef MH. Occurrence and predictors of dexmedetomidine infusion intolerance and failure. *Hosp Pract (1995).* 2012;40(1):186-192. doi:10.3810/hp.2012.02.959 [PubMed 22406894]
160. Thibault C, Moorthy GS, Vedar C, Naim MY, DiLiberto MA, Zuppa AF. Pharmacokinetics of cefepime in children on extracorporeal membrane oxygenation: external model validation, model improvement and dose optimization. *Pediatr Infect Dis J.* 2022;41(3):217-223. doi:10.1097/INF.0000000000003371 [PubMed 34817416]
161. Tobias JD. Dexmedetomidine: Are there going to be issues with prolonged administration? *J Pediatr Pharmacol Ther.* 2010;15(1):4-9. [PubMed 22477787]
162. Tobias JD, Berkenbosch JW. Sedation during mechanical ventilation in infants and children: dexmedetomidine versus midazolam. *South Med J.* 2004;97(5):451-455. [PubMed 15180019]
163. US Food and Drug Administration; Institute for Safe Medication Practices. FDA and ISMP Lists of Look-Alike Drug Names With Recommended Tall Man (Mixed Case) Letters. ISMP; 2023.
164. Venn RM, Karol MD, Grounds RM. Pharmacokinetics of dexmedetomidine infusions for sedation of postoperative patients requiring intensive care. *Br J Anaesth.* 2002;88(5):669-675. [PubMed 12067004]
165. Venn M, Newman J, Grounds M. A phase II study to evaluate the efficacy of dexmedetomidine for sedation in the medical intensive care unit. *Intensive Care Med.* 2003;29(2):201-207. doi:10.1007/s00134-002-1579-9 [PubMed 12594584]
166. Vilo S, Rautiainen P, Kaisti K, et al. Pharmacokinetics of intravenous dexmedetomidine in children under 11 yr of age. *Br J Anaesth.* 2008;100(5):697-700. [PubMed 18378546]
167. Walker J, Maccalum M, Fischer C, et al. Sedation using dexmedetomidine in pediatric burn patients. *J Burn Care Res.* 2006;27(2):206-210. [PubMed 16566567]
168. Wang C, Liu S, Han C, Yu M, Hu Y, Liu C. Effect and placental transfer of dexmedetomidine during caesarean section under epidural anaesthesia. *J Int Med Res.* 2017;45(3):964-972. doi:10.1177/0300060517698330 [PubMed 28449631]
169. Wang SS, Zhang MZ, Sun Y, et al. The sedative effects and the attenuation of cardiovascular and arousal responses during anesthesia induction and intubation in pediatric patients: a randomized comparison between two different doses of preoperative intranasal dexmedetomidine. *Paediatr Anaesth.* 2014;24(3):275-281. [PubMed 24224515]
170. Wang T, Zhang H, Tian R, et al. The effect of extracorporeal membrane oxygenation on the pharmacokinetics of dexmedetomidine hydrochloride. *J Cardiothorac Vasc Anesth.* 2025;39(3):616-624. doi:10.1053/j.jvca.2024.12.029 [PubMed 39779430]
171. Wang Y, Fang X, Liu C, Ma X, Song Y, Yan M. Impact of intraoperative infusion and postoperative PCIA of dexmedetomidine on early breastfeeding after elective cesarean section: a randomized double-blind controlled trial. *Drug Des Devel Ther.* 2020;14:1083-1093. doi:10.2147/DDDT.S241153 [PubMed 32210537]
172. Wang YQ, Zhang XJ, Wang Y. Effect of intrathecal dexmedetomidine on cesarean section during spinal anesthesia: a meta-analysis of randomized trials. *Drug Des Devel Ther.* 2019;13:2933-2939. doi:10.2147/DDDT.S207812 [PubMed 31686777]
173. Weber MD, Thammasitboon S, Rosen DA. Acute discontinuation syndrome from dexmedetomidine after protracted use in a pediatric patient. *Paediatr Anaesth.* 2008;18(1):87-88. doi:10.1111/j.1460-9592.2007.02377.x [PubMed 18095980]

174. Weerink MAS, Struys MMRF, Hannivoort LN, et al. Clinical pharmacokinetics and pharmacodynamics of dexmedetomidine. *Clin Pharmacokinet.* 2017;56(8):893-913. doi:10.1007/s40262-017-0507-7 [PubMed 28105598]
175. Weiner MM, Chow R, Salter BS. Case report: a case of fetal bradycardia following dexmedetomidine bolus. *Journal of Obstetric Anaesthesia and Critical Care.* 2014;4(2):75-77. doi:10.4103/2249-4472.143876
176. Whalen LD, Di Gennaro JL, Irby GA, Yanay O, Zimmerman JJ. Long-term dexmedetomidine use and safety profile among critically ill children and neonates. *Pediatr Crit Care Med.* 2014;15(8):706-714. doi:10.1097/PCC.0000000000000200 [PubMed 25068249]
177. Wu J, Mahmoud M, Schmitt M, Hossain M, Kurth D. Comparison of propofol and dexmedetomidine techniques in children undergoing magnetic resonance imaging. *Paediatr Anaesth.* 2014;24(8):813-818. doi:10.1111/pan.12408 [PubMed 24814202]
178. Xu A, Wan L. Dexmedetomidine-induced polyuric syndrome and hypotension. *J Clin Anesth.* 2018;44:8-9. doi:10.1016/j.jclinane.2017.10.008 [PubMed 29078065]
179. Xu B, Zhou D, Ren L, Shulman S, Zhang X, Xiong M. Pharmacokinetic and pharmacodynamics of intravenous dexmedetomidine in morbidly obese patients undergoing laparoscopic surgery. *J Anesth.* 2017;31(6):813-820. doi:10.1007/s00540-017-2399-y [PubMed 28828532]
180. Yamasaki K, Tokuno M, Tsukigawa K, et al. Possible involvement of protein binding inhibition in changes in dexmedetomidine concentration in extracorporeal circuits during midazolam use. *Biol Pharm Bull.* 2024;47(2):389-393. doi:10.1248/bpb.b23-00659 [PubMed 38325827]
181. Yoshimura M, Kunisawa T, Suno M, et al. Intravenous dexmedetomidine for cesarean delivery and its concentration in colostrum. *Int J Obstet Anesth.* 2017;32:28-32. doi:10.1016/j.ijoa.2017.05.002 [PubMed 28687146]
182. Yu M, Han C, Jiang X, Wu X, Yu L, Ding Z. Effect and placental transfer of dexmedetomidine during caesarean section under general anaesthesia. *Basic Clin Pharmacol Toxicol.* 2015;117(3):204-208. doi:10.1111/bcpt.12389 [PubMed 25652672]
183. Yu Q, Liu Y, Sun M, et al. Median effective dose of intranasal dexmedetomidine sedation for transthoracic echocardiography in pediatric patients with noncyanotic congenital heart disease: An up-and-down sequential allocation trial. *Paediatr Anaesth.* 2017;27(11):1108-1114. doi:10.1111/pan.13235 [PubMed 28940686]
184. Yuen VM, Cheuk DK, Hui TW, Wong IC, Lam WW, Irwin MG. Oral chloral hydrate versus intranasal dexmedetomidine for sedation of children undergoing computed tomography: a multicentre study. *Hong Kong Med J.* 2019;25 Suppl 3(1):27-29. [PubMed 30792370]
185. Yuen VM, Hui TW, Irwin MG, et al. A randomised comparison of two intranasal dexmedetomidine doses for premedication in children. *Anaesthesia.* 2012;67(11):1210-1216. [PubMed 22950484]
186. Yuen VM, Irwin MG, Hui TW, Yuen MK, Lee LH. A double-blind, crossover assessment of the sedative and analgesic effects of intranasal dexmedetomidine. *Anesth Analg.* 2007;105(2):374-380. [PubMed 17646493]
187. Zapantis A, Leung S. Tolerance and withdrawal issues with sedation. *Crit Care Nurs Clin North Am.* 2005;17(3):211-23. doi:10.1016/j.ccell.2005.04.011 [PubMed 16115529]
188. Zhang J, Zhou H, Sheng K, Tian T, Wu A. Foetal responses to dexmedetomidine in parturients undergoing caesarean section: a systematic review and meta-analysis. *J Int Med Res.* 2017;45(5):1613-1625. doi:10.1177/0300060517707113 [PubMed 28521658]
189. Zhang SY, Zhao H, Xu C, et al. Combination of dexmedetomidine and tramadol in patient-controlled intravenous analgesia strengthens sedative effect in pregnancy-induced hypertension. *Front Pharmacol.* 2021;12:739749. doi:10.3389/fphar.2021.739749 [PubMed 34744722]

190. Zhang X, Schmidt U, Wain JC, Bigatello L. Bradycardia leading to asystole during dexmedetomidine infusion in an 18 year-old double-lung transplant recipient. *J Clin Anesth.* 2010;22(1):45-49. doi:10.1016/j.jclinane.2009.06.002 [PubMed 20206851]
191. Zhang X, Wang R, Lu J, et al. Effects of different doses of dexmedetomidine on heart rate and blood pressure in intensive care unit patients. *Exp Ther Med.* 2016;11(1):360-366. doi:10.3892/etm.2015.2872 [PubMed 26889269]
192. Zhang X, Zhao X, Wang Y. Dexmedetomidine: a review of applications for cardiac surgery during perioperative period. *J Anesth.* 2015;29(1):102-111. doi:10.1007/s00540-014-1857-z [PubMed 24913070]
193. Zhu J, Lu D, Liu JF. Intraoperative dexmedetomidine-related polyuria: a case report and review of the literature. *Int J Clin Pharmacol Ther.* 2022;60(4):188-191. doi:10.5414/CP204129 [PubMed 34889734]
194. Zub D, Berkenbosch JW, Tobias JD. Preliminary experience with oral dexmedetomidine for procedural and anesthetic premedication. *Paediatr Anaesth.* 2005;15(11):932-938. doi:10.1111/j.1460-9592.2005.01623.x [PubMed 16238552]

Topic 8897 Version 729.0

Source: UpToDate. Downloaded 2026-05-26.